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 SURVEY

Electronic Warfare Cyberattacks, Countermeasures, and Modern Defensive Strategies of UAV Avionics: A Survey

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ABSTRACT Unmanned Aerial Vehicles (UAVs) play a pivotal role in modern autonomous air mobility, and the reliability of UAV avionics systems is critical to ensuring mission success, sustainability practices, and public safety. The success of UAV missions depends on effectively mitigating various aspects of electronic warfare, including non-destructive and destructive cyberattacks, transponder vulnerabilities, and jamming threats, while rigorously implementing countermeasures and defensive aids. This paper provides a comprehensive review of UAV cyberattacks, countermeasures, and defensive strategies. It explores UAV-to-UAV coordination attacks and their associated features, such as dispatch system attacks, Automatic Dependent Surveillance-Broadcast (ADS-B) attacks, Traffic Alert and Collision Avoidance System (TCAS)-induced collisions, and TCAS attacks. Additionally, the paper examines UAV-to-command center coordination attacks, as well as UAV functionality attacks. The review also covers various countermeasures and defensive aids designed for UAVs. Lastly, a comparison of common cyberattacks and countermeasure approaches is conducted, along with a discussion of future trends in the field.

INDEX TERMS Electronic warfare, UAVs, avionics systems, cyberattacks, coordination attacks, functionality attacks, countermeasure, defensive-aids.

I. INTRODUCTION

A. MOTIVATION

Unmanned Aerial Vehicles (UAV) have become increasingly important in both civilian and military applications over the past several decades. Within the civil sector, UAVs have seen significant growth, finding use in news broadcasting, agriculture, construction, and other business ventures [1], [2], [3], [4]. Military UAVs have also been used extensively in surveillance and object targeting [1], [5]. These prospects are made possible by the major trends in UAV development, with current research targeting increasing levels of autonomy

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through advancements in on-board avionics systems [6], [7]. The general UAV avionics composition is similar to those found on manned aircrafts and can be broken into several subsystems: propulsion, electrical, sensors, communications, flight navigation, and control [8]. The avionics system interacts with these subsystems for different purposes, such as controlling and maintaining flight altitude with the propulsion system, drawing power from the batteries of the electrical system, or collecting data from the sensor system. The integration and control of these processes is a complex task that often relies on one or more flight computers [1], [7], [8]. Communications between UAVs (air-to-air) and to ground stations (air-to-ground) is a critical component of UAV networks, as it provides the proper means

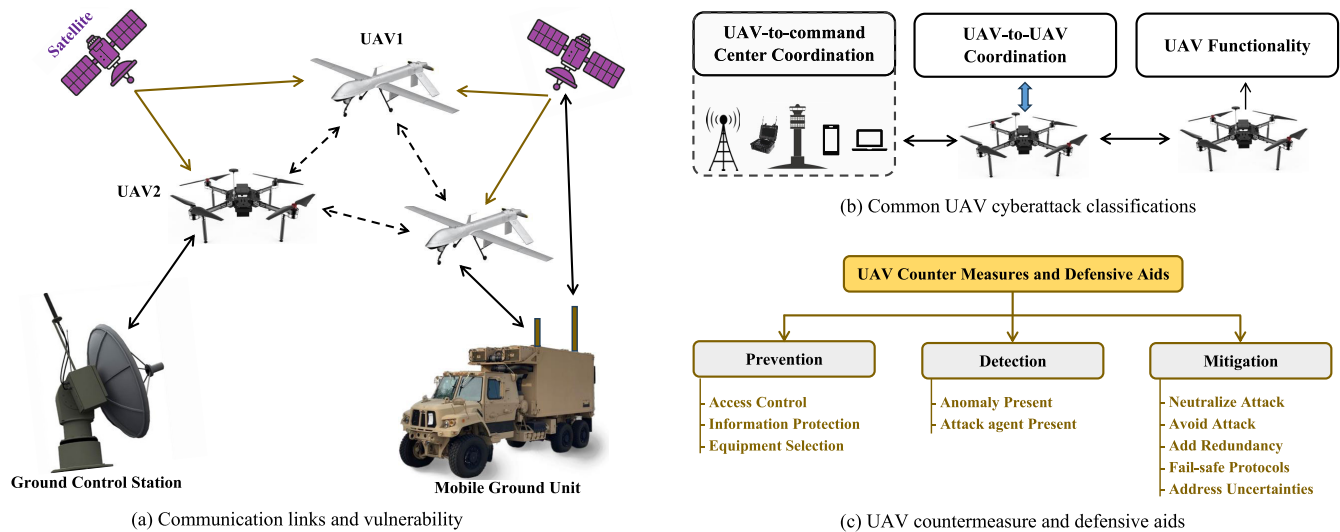


FIGURE 1. Conceptual illustration of UAV electronic warfare: (a) Communication links, (b) common UAV cyberattack classifications, and (c) countermeasures and defensive aids.

of identification and communication for drones to accomplish their individual tasks. Some of the commonly used notation are listed in Table 1.

B. ELECTRONIC WARFARE AND UAVs

Electronic warfare encompasses the science and art of maintaining control over the Electromagnetic Spectrum (EMS) for friendly use while denying its use to adversaries. In the context of UAVs, this primarily involves military tactics, such as information gathering and disruption of operations by exploiting communication links within UAV networks, as illustrated in Figure 1(a). This area of warfare is an expanding field of research, with a growing focus on developing strategies and techniques that can both protect friendly UAVs from hostile actors and disrupt the operations of adversarial UAVs. For civilian UAVs, this type of warfare is more commonly referred to in broader scopes in relation to cybersecurity and cyberattacks [9], [10]. Along with UAVs, electronic warfare began development during World War II and picked up significantly during following conflicts. Jamming techniques were first explored in 1902 by the Royal Navy with the introduction of the radio. They used for the first time in combat by a Russian operator during the 1905 Russo-Japanese war by overlapping signals meant for the correction of Japanese naval gunfire [11]. Once RADAR was established in England, 1935, further research into jamming began along with the official introduction of electronic warfare. This led to the development of RADAR reflectors used in 1943 to confuse German RADAR operators and win British victories. They were used again in 1945 to completely hide U.S. aircraft from Japanese searchlight RADARs [12]. However, post-war electronic warfare development halted for a half-decade as engineers returned to civilian life and most military equipment were sold off. No U.S. study on lessons-learned for electronic warfare was produced, and

thus most successes and developments were never fully documented. The following conflicts involving the U.S. were limited in their electronic warfare capabilities due to this gap in knowledge [11]. The extent of electronic warfare in the Korean War was the U.S. B-12s jamming Chinese RADAR systems. Nevertheless, advancements in UAV technology and electronic warfare have continued into the twenty-first century, resulting in modernized capabilities.

C. CYBERATTACKS RELATED TO UAVs

Due to the increasing use of UAVs for commercial applications, cyberattacks are a concern for both military and civilian class drones [13]. Various classifications of UAV related cyberattacks exist, with some examples being destructive and non-destructive attacks [14], ground-based and air-based attacks [15], and threat categorized attacks ranging from physical, sensor, and communication, to supply chain threats [16]. The classifications can then further be characterized by their military impact as opposed to civilian impacts. Because of the rapidly growing development of UAVs in the public sector, most papers tend to focus on the civilian side of UAV cyberattacks and their repercussions. Drone cyberattacks can be split into three classes: UAV-to-UAV coordination attacks, UAV-to-command center coordination attacks, and UAV functionality attacks. In civil applications, drones are often deployed in groups to carry out tasks. Even singular UAVs typically share the air space with other aerial vehicles, which can include manned aircraft alongside other drones on separate missions. As a result, UAV-to-UAV coordination attacks involve interfering in the messaging systems amongst UAVs. In serious cases, such attacks can lead to midair collisions among other issues [17]. Commonly termed attacks that belong to this category include dispatch system attacks, Automatic Dependent

TABLE 1. Nomenclature.

ADS-B	Automatic Dependent Surveillance - Broadcast
ATC	Air Traffic Control
COMINT	Communications Intelligence
DoS	Denial-of-Service
DSSS	Direct Sequence Spread Spectrum
EA	Electronic Attack
EASA	European Union Aviation Safety Agency
ELINT	Electronic Intelligence
EMS	Electromagnetic Spectrum
EP	Electronic Protection
ES	Electronic Warfare Support
FAA	Federal Aviation Administration
FH	Frequency Hopping
FHSS	Frequency Hopping Spread Spectrum
FL	Federated Learning
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
IFF	Identify Friend or Foe
IMU	Inertial Measurement Unit
ICAO	International Civil Aviation Organization
LiDAR	Light Detection and Ranging
LTE	Long-Term Evolution
MAC	Media Access Control
NMAC	Near Midair Collision
QKD	Quantum Key Distribution
RID	Remote Identification
SATCOM	Satellite Communications
SDR	Software-defined Radio
SIGINT	Signal Intelligence
TCAS	Traffic Alert and Collision Avoidance System
TCP/UDP	Control Protocol/User Datagram Protocol
UWB	Ultra-wideband
UAVs	Unmanned Aerial vehicles

Surveillance - Broadcast (ADS-B) attacks, and Traffic Alert and Collision Avoidance System (TCAS) induced collisions. In a UAV network, the command center is responsible for keeping track of and maintaining each UAV's operational status such as payload control, mission planning, and air vehicle control [4]. This information is transmitted to the UAV through wireless communication links that, because of historical development in transponder technology, is not inherently secure. Cyberattacks of this nature passively or actively disrupt the communication link between the UAV and command center. Passive attacks, such as eavesdropping, focus on obtaining information from the communication link for malicious purposes, while active attacks such as jamming actively interfere with the UAV's ability to communicate to the command center. Unlike the previous two categorizations, cyberattacks targeting the UAV's functionality precisely alter the way the UAV behaves as opposed to interfering with its communication channels [17], [18], [19], although the methods of access can be similar. This is dependent on the particular UAV's hardware configuration; however common

examples include exploitation of recorded video attacks and Global Positioning System (GPS) spoofing. In such attacks, fake signals are generated by malicious actors which are then sent to the UAV. The UAV believes that the fake information is genuine, altering the way it executes its mission depending on the type of attack being performed. In many cases, this attack tricks the UAV into believing its spoofed location is legitimate, allowing attackers to hijack the UAV's flight path and steal it. Figure 1.(b) shows categorization of various cyberattacks on UAVs exposed by UAV systems in public as well as military drones.

D. COUNTERMEASURES AND DEFENSIVE-AIDS

Electronic countermeasure systems employ specific tactics, techniques, and technologies to interfere with, deceive, disrupt, degrade, or neutralize an adversary's electronic and radar systems of the enemy [20], [21], [22]. Electronic counter-countermeasure systems, as part of defensive aid measures, are designed to counteract electronic countermeasures by restoring radar functionality and mitigating their effects [23]. Additionally, electronic counter-countermeasure systems encompass strategies to protect friendly systems from electronic threats, such as encryption and signal shielding. Electronic countermeasure and electronic counter-countermeasure are known as cat and mouse game. The most common threats faced by aircraft are small arms fire, radar guided anti-aircraft missiles, shoulder-launched surface-to-air missiles (SAM), and SAM mounted to ground sites, vehicles or ships [24]. On the other side, the defensive-aid subsystem related to aircrafts would include radar warning receiver, missile warning receiver, laser warning receiver, countermeasure dispenser (chaff or flares), and towed decoy [25]. With regard to UAVs, countermeasures and defensive-aids are methods applied to protect the data handled by UAVs, more specifically, confidentiality, integrity, and authenticity [21], [22], [26], [27], [28], [29], [30], [31]. Also, countermeasures and defensive-aids for UAVs are used to ensure the service availability for civilian and military applications [32]. In view of UAV cyberattacks, a wide spectrum of countermeasures are created to take advantage of vulnerabilities found on hardware, software, and network layers [1], [33]. The same countermeasure may work against several types of attacks covered under three broad categories: prevention, detection, and mitigation (as shown in Figure 1.(c)).

E. OTHER IMPACTS OF ELECTRONIC WARFARE

Electronic warfare often involves the use of high-powered electronic signals, which can contribute to electromagnetic pollution, potentially harming the environment and living organisms, and negatively impacting sustainability efforts [32]. High levels of electromagnetic radiation may interfere with the navigation systems of wildlife, leading to behavioral disruptions. In terms of energy consumption, electronic warfare systems, such as jammers, radars, and

signal intelligence receivers, demand substantial energy to operate. These systems frequently rely on portable generators or ground/aerial vehicles, resulting in increased fuel consumption, greenhouse gas emissions, and subsequent environmental degradation. Additionally, electronic warfare can disrupt civilian communication and navigation systems, including emergency services and GPS-based systems [4]. This disruption can lead to an increased risk of accidents in sensitive areas and a reduction in the effectiveness of environmental monitoring and response systems.

F. FOCUS AND STRUCTURE

1) SCOPE

This survey article aims to provide a comprehensive overview of UAV electronic warfare, encompassing cyberattacks, countermeasures, and defensive aids. It offers a detailed characterization of various attack taxonomies and systematically reviews UAV-to-UAV coordination attacks, classifying them into distinct categories such as dispatch system attacks (including message elimination, message spoofing, and message fabrication), ADS-B attacks, TCAS-induced collisions, and TCAS attacks. Additionally, UAV-to-command center coordination attacks are analyzed, covering threats such as eavesdropping, man-in-the-middle attacks, jamming, and Wi-Fi-based intrusions. UAV functionality attacks are also explored, addressing risks such as recorded video exploitation, Denial-of-Service (DoS) attacks, and GPS spoofing. Beyond categorizing cyber threats, this article examines UAV electronic warfare in both civil and military contexts, discussing countermeasures and defensive aids related to prevention techniques, information security, communication traffic management, mitigation strategies, and the avoidance of wireless communication vulnerabilities. Furthermore, a comparative analysis of common cyberattacks and countermeasure approaches is presented, along with an exploration of emerging trends in the field. Unlike prior works such as [20], [34], and [35], which focus on network attacks and the corresponding countermeasures for communication signals, this study is dedicated specifically to UAV electronic warfare, including cyberattacks, countermeasures, and modern defensive-aid techniques. The studies in [36], [37], and [38] primarily address identification threats but do not extensively examine UAV-to-UAV coordination attacks, UAV-to-command center coordination attacks, or functionality-based attacks. Similarly, [39] investigates various types of UAV communication attacks but lacks an in-depth discussion of countermeasures and defensive-aid strategies. Furthermore, the studies in [40] and [41] emphasize communication network security, software security analysis, and intelligent security, with a primary focus on functionality attacks. However, they do not comprehensively explore the role of transponders in security threats, the distinctions between security challenges in manned aircraft and UAVs, or UAV-to-UAV coordination attacks. The works in [42] and [43] are primarily structured around cryptographic methods for

securing communications and developing security protocols. Therefore, this study provides a more extensive and holistic discussion of UAV-to-UAV coordination attacks including those involving transponders UAV-to-command center coordination attacks, UAV functionality attacks, and their associated countermeasures and defensive aids. Additionally, it presents an analysis of current challenges and outlines future research directions in the domain of UAV electronic warfare.

2) STRUCTURE

The rest of the paper is composed of eight sections. Section II presents UAV-to-UAV coordination attacks. Section III discusses UAV-to-command center coordination attacks. Section IV summarizes UAV functionality attacks. Section V discusses UAV electronic warfare for military purposes. Section VI presents countermeasures and popular defensive-aids. Section VII provides comparison of UAV cyberattacks and countermeasures strategies. Section VIII presents future trends. Finally, Section IX concludes the work.

II. UAV-TO-UAV COORDINATION ATTACKS

A. TRANSPONDERS OVERVIEW

The purpose of Identify Friend or Foe (IFF) is to identify friendly and nonfriendly aircraft as the number of aircraft in airspace increases. For comprehensive overview visit [32]. Generally, this consists of an airborne receiver that listened for primary radar transmissions and would reply, at the same frequency, with a message specific to that aircraft [44]. TCAS is an avionics system that relies on transponders to perform air-to-air interrogation and warns the pilot about dangerous encounters with nearby aircraft [45]. This is a cooperative system, meaning that for it to function all aircraft should be equipped with a mode S transponder to perform surveillance (Mode S surveillance). Mode C surveillance can also be performed by TCAS to interrogate aircraft with Mode A/C transponders [46]. TCAS sends out pulses and receives responses from neighboring aircraft transponders which contain range, altitude, and azimuth (bearing). When the system detects other aircraft that are at safe range and altitude away the pulses are sent once a minute to perform passive surveillance. Passive surveillance continues when the spacecraft is at either close range or altitude at higher rate of once per 10 seconds. Once the aircraft enters potential collision zone (i.e. it is close in both altitude and range) TCAS engages in active surveillance at rate of 1 pulse per second [46]. ADS-B is another transponder-based avionics system that aids aircraft surveillance. Unlike IFF and TCAS it does not rely on interrogations from either Air Traffic Control (ATC) or other aircraft as it automatically broadcasts flight state parameters to everyone involved in the network [47]. Similar to TCAS, ADS-B is a cooperative sensing system, meaning that in order to receive ADS-B broadcast all players must have ADS-B system on board. ADS-B uses Mode S transponder and broadcasts identification, position,

and velocity at frequency of 1 Hz. Position and velocity are determined from GNSS signals in conjunction with data from sensor systems (Inertial Measurement Unit (IMU), barometer, etc.).

Since transponders converse by broadcasting information encoded in electromagnetic waves, it is not surprising to believe that the electromagnetic environment surrounding drones can become densely proliferated by noise leading to communication issues. However, with over 400,000 expected drones on daily basis for use in commercial and government missions in European airspace by 2050 [48], the matter of bandwidth limitations between similarly broadcasting UAVs becomes an important topic of concern [49]. Evidently, the cumulation of messages from increasing numbers of sources results in an electromagnetic environment indiscernible from noise. For instance, ADS-B congestion of the 1090 MHz band caused the Federal Aviation Administration (FAA) to prohibit the use of ADS-B on UAVs [49], which in turn has prompted research into alternative ADS-B “like” technologies to accommodate the requirements of larger drone networks [49], [50]. European Union Aviation Safety Agency (EASA) shares the same concern with EMS usage and safety [7]. When network congestion occurs, transponders become more likely to fail their tasks since interference limits their ability to decode interrogations from other UAVs (air-to-air) or ground stations (air-to-ground).

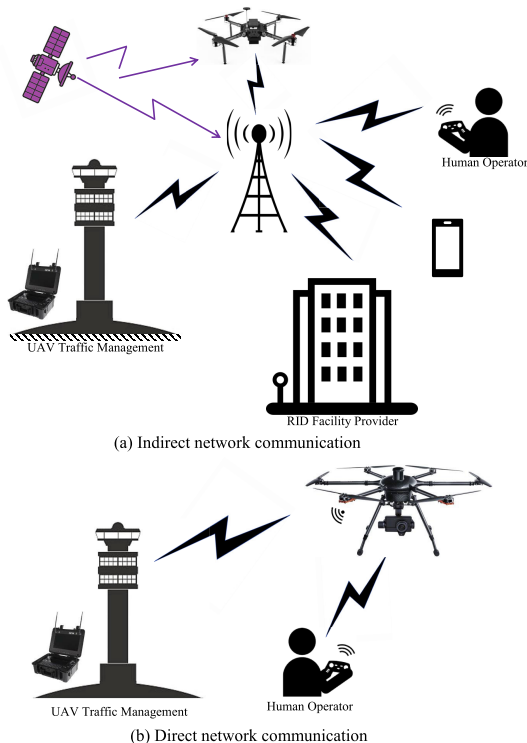


FIGURE 2. RID Communication Technologies.

While remote identification (RID) is a general term, in the context of drones, it refers to the larger group of new methods of identification for UAVs specifically that has

not been historically associated with aviation [32]. While ADS-B, TCAS and IFF have been tried on aircraft, alternative methods may prove more beneficial for UAVs. There are two categories of remote identification [36]. First are technologies that continuously broadcast the location of the drone to local receivers. This is done by having a GPS receiver on the drone, and a transmitter on the drone that sends out over Wi-Fi, Bluetooth or LoRa frequencies. The second category is network communication, which uses a cellular network that the operator must subscribe the drone to as one would for a cellular phone. The broadcasting method is mainly intended for short range and typically only includes communication from the drone to only the operator. The appeal of these broadcasting methods is that the relevant technology does not require a licence to purchase or operate. It would be up to the operator’s receiving device to then transmit the information to other parties, such as ATC or other aircraft/UAVs. Fig. 2 presents RID communication technologies. Using Wi-Fi as the connection can work up to 2 km if using neighbor awareness networking and consumes 100 mW from the drone power system. However, this has yet to be implemented on a drone. The DJI drone company has attempted to use lower range Wi-Fi to track many drones in an area by transmitting and receiving directly from the drone to mobile phones. This has worked up to about 1 km. For ATC tracking of drones, UAV-to-UAV, or UAV to aircraft identification, that information would have to be recommunicated by the operator’s mobile device. Another method, Bluetooth, has low bandwidth that can reach up to 200 m if using Bluetooth 4.x or 1 km if using Bluetooth 5.x, consuming a maximum of 10 mW. Unfly company’s Broadcast Location & Identification Platform (BLIP) uses Bluetooth to broadcast drone details for up to 200 m and transfer that data through operator device’s Long-Term Evolution (LTE) network connection. LTE is associated to fourth-generation (4G). LoRa can reach a longer distance with a higher bit rate than Wi-Fi or Bluetooth [36], [51]. Considering the discussion above Table 2 goes into detail about how each of the transponders compare for implementation on drones.

B. DISPATCH SYSTEM ATTACKS

A dispatch system refers to the hardware and software protocols and algorithms devoted to ensuring that a group of drones with a common objective can autonomously perform their tasks without impeding one another [52]. Consequently, any attack on the dispatch system can possibly alter the behaviour of all the drones on the network. The most common way to perform this attack is by injecting malware into the system [17]. The malware typically includes malicious firmware at the software level, and trojans at the hardware level [53], which are injected to produce vulnerabilities within the UAV’s flight controller as well as the ground control station. Software level malware can affect the UAV in countless ways that are not limited to dispatch system attacks. An example that falls under this category of attack is the

TABLE 2. Transponder comparisons in the context of implementation on drones.

Transponder	Weight	Power	Access	Advantages	Disadvantages to ATC
IFF/SSR	Heavy	High	Low	Has been attempted	Requires radio license and yet to be implemented
TCAS	Heavy	High	Medium	Has been attempted	Only implemented within research context
ADS-B	Heavy	High	Medium	Implemented in academic research	Only implemented within research context
RID: Wi-Fi	Light	Low	High	Easy to obtain and use, no license	Short range, 2km, cannot fly drone very far
RID: Bluetooth	Light	Low	High	Easy to obtain and use, no license	Very short range, 200m
RID: LoRa	Light	Low	Medium	Prioritizes operator control	Requires radio license and interference probability
RID: Cellular Network	Light	Low	Medium	More affordable and easier to implement than ADS-B	Infrastructure not ready for global coverage

Maldrone virus [54]. Maldrone was developed as the first ever backdoor malware written for AR drone ARM Linux systems. It allows a remote hijacker to remotely switch the drone's software to obey their backdoor controller, providing remote manipulation and access to the drone [54]. Evidently, such viruses can be devastating for a network of drones if even one such drone becomes infected by a malicious actor. Hardware trojans, on the other hand, include hardware level modifications to the circuit of the flight controller. Due to the complex underlying systems that comprise the flight controller, trojans are most commonly introduced due to non-trusted, imitation hardware being used at some point in the supply chain. These security breaches compromise the functions of the circuit itself, leading to premature failure of components and untimely destruction of the drone [53]. In more severe cases, the trojan can introduce backdoors that leak information to the attackers or allow total takeover of the drone itself. An example of such a trojan was a keylogging virus installed in a ground control unit of the U.S. Air Force, leading to a backdoor access which malicious actors used to track the keystrokes made in controlling the U.S. drone fleet over Iraq and Afghanistan [37].

C. ADS-B ATTACK

ADS-B transponders are used in drones to provide autonomous collision avoidance capabilities, which becomes more important with increasing numbers of drones in the same air space. Attacks on the ADS-B system involve exploiting the fact that transmitted messages are sent in plain text format [17], and do not inherently provide any authentication methods to stop message tampering [55], [56]. These attacks can further be characterized into the following three categories: Message elimination, message infusion (spoofing), and message fabrication.

1) MESSAGE ELIMINATION

It involves using external transmitters to project constructive or destructive interference into the ADS-B signal. When

constructive interference is used, the attack induces bit errors into the ADS-B message, causing the receiving drones to disregard the message once it detects the manipulation and thus diminishing awareness of the transmitter drone. When destructive interference is used, the attacking signal is an inverse of the original ADS-B signal, leading to complete or partial destruction of the message [53].

2) MESSAGE INFUSION (SPOOFING)

The approach of message infusion involves injecting malicious messages into the airspace, causing ADS-B receivers to perceive the appearance of an illegitimate aircraft. This is possible since ADS-B does not use any authentication methods in its messages, and so infusion can be performed by commercially available devices. The broadcasted false messages can either target the UAVs themselves (Aircraft Target Ghost Injection) or the ground command center (Command Center Ghost Injection). In both cases, the target ADS-B receiver sees a fake aircraft in the air space as the attacker anonymously manipulates the air traffic [52]. ADS-B spoofing is dangerous as it can allow enemies to masquerade as potential allies in the view of the ground station. Figure 3 shows how a spoofing situation might look, where the presence of ground-based and aircraft-based attackers infiltrate the airspace of authentic aircraft, causing the ground station to perceive 3 different and friendly aircraft.

3) MESSAGE FABRICATION

Message fabrication involves manipulation of ADS-B signals to provide false information which ADS-B receivers then interpret. Unlike message infusion, message fabrication manipulates the messages sent by legitimate UAVs. The level of tampering can vary based on the intention. Overshadowing is a method where the attacker broadcasts a very high-powered ADS-B signal to substitute parts of the ADS-B message, or entire message in the worst case. Similarly, bit flipping involves flipping certain bits to partially manipulate

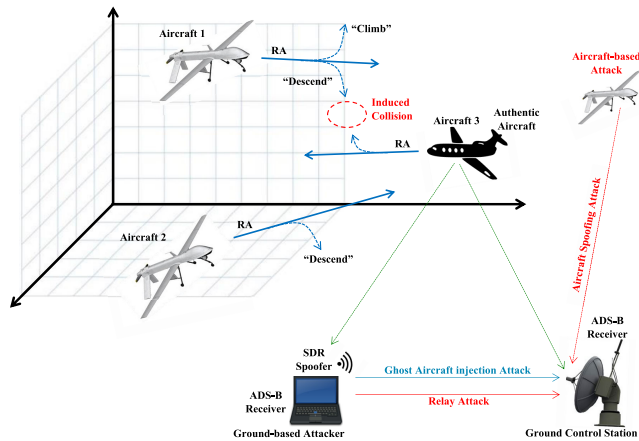


FIGURE 3. Illustration of TCAS induced collision example and ADS-B spoofing attacks through a ground-based attack, which replays a recorded ADS-B signal, and an aircraft-based attack, which spoofs the aircraft's ICAO address.

the signal. In either case, information is removed from the original message and malicious data is inserted in its place [17]. These ADS-B attacks can be accomplished using commercial off-the-shelf components as well as freely licensed software [57]. The most popular example of this is with Software-defined Radio (SDR) devices, which can be programmed to transmit radio signals of different frequencies, including those of ADS-B signals. By following the general packet structure of an ADS-B message, Figure 3, any SDR can effectively mask themselves as an ADS-B transmitter.

D. TCAS INDUCED COLLISION

Similar to ADS-B, TCAS is used to coordinate maneuvers between UAVs in a local air space for the purpose of avoiding collisions. However, the resolution advisories produced by TCAS do not predict long term effects. This leads to the problem of TCAS induced collision, wherein the resolution advisories provided by TCAS result in a collision. Figure 3 shows an example of such a situation [17]. Aircraft pairs 1 and 2, as well as 3 and 4, are originally on collision paths but are provided resolution advisories from TCAS. Since the TCAS advisories were made independently, this can lead to a situation where aircraft 4 climbs while aircraft 1 descends, leading to an induced collision with less time to provide corrections as a consequence of TCAS decisions. This can be extended to the case of UAVs, wherein an attacker tampers with air traffic data to cause a TCAS induced collision [52].

E. TCAS ATTACK

Despite TCAS strong reputation regarding safety, it was not designed to withstand any type of attacks, i.e. jamming [47]. One of the general ways is to jam 1090 MHz channel to prevent the aircraft from tracking potential "intruders". However, these types of attacks are easily detected and

countered [47], [58]. More effective jamming attacks include so called "All-Call Flood" and "Squitter Flood". During the "All-Call Flood" the attacker takes advantage of All-call interrogation and occupies 1030 MHz channel to trigger all nearby Mode S transponders to reply with their 24-bit International Civil Aviation Organization (ICAO) address and flood the 1090 MHz reply channel. The "Squitter Flood" attack is performed by an attacker spoofing the nearby transponders by transmitting replies on 1090 MHz channel and forcing them to continuously track the "false" aircraft. Both of these attacks increase the chances of Near Midair Collision (NMAC) events however the attacker does not have full control over the NMAC occurrence. Different kind of attack that is not related to channel flooding is called "Phantom Aircraft" attack. If the attacker can produce an accurate Mode S reply and seem to move like an airplane, the TCAS transponder assumes that these replies are coming from an actual aircraft, which will force it into tracking. Such false tracking can lead to generation of RAs that could lead to an NMAC [47].

F. UAV VS MANNED AIRCRAFTS

Traditional aircraft rely on secure and redundant avionics systems, including protected ATC channels and hardened cockpit systems, which reduce their susceptibility to cyberattacks [6], [59], [60]. In contrast, UAVs depend on unmanned control links, such as SATCOM, Wi-Fi, 4G/5G, or RF signals, making them highly vulnerable to jamming, signal spoofing, and interception. A compromised control link in UAVs can result in hijacking (e.g., GPS spoofing) or complete mission failure [18], [61]. Manned aircraft incorporate hardened avionics systems, including shielded onboard computers, encrypted flight control software, and physically secured cockpits, which mitigate the impact of cyber intrusions. In contrast, UAVs particularly commercial and consumer-grade drones often lack secure hardware architectures and may rely on open-source or commercially available flight controllers [62], [63]. This lack of security makes them more susceptible to firmware exploitation, malware injection, and unauthorized software modifications. UAVs are inherently more vulnerable than manned aircraft due to several factors, including the absence of human intervention, heavy reliance on wireless communication, and the presence of unhardened systems in commercial drones, which often lack robust cybersecurity measures. Furthermore, autonomous decision-making introduces additional risks, such as adversarial AI attacks that could disrupt UAV navigation and mission planning, as well as vulnerabilities in swarm coordination that may compromise entire fleets. As UAV adoption increases, enhancing their cyber resilience is essential for ensuring mission security in both civilian and military applications. This can be achieved through the implementation of secure AI, encrypted communication protocols, anti-jamming mechanisms, and blockchain-based authentication systems.

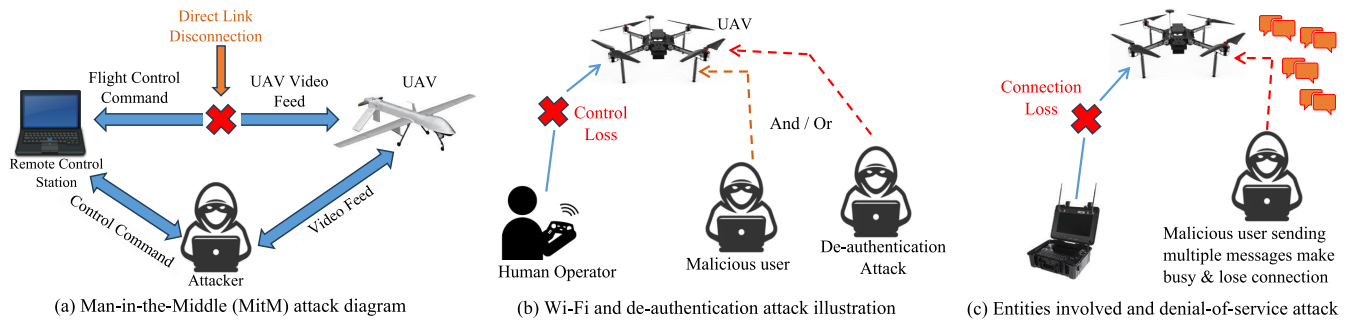


FIGURE 4. UAV-to-command center coordination attacks:(a) man-in-the-middle attack; (b) Wi-Fi attack and/or de-authentication attack, wherein a malicious attacker closes the communication channel used by a legitimate user; and (c) Entities involved in a DoS attack.

III. UAV-TO-COMMAND CENTER COORDINATION ATTACKS

A. EAVESDROPPING ATTACK

It is a passive yet highly insidious form of cyber threat in UAV networks. In this type of attack, malicious actors clandestinely listen to and record unencrypted data transmissions between UAVs and their ground control centers. While this attack does not directly disrupt the operation or functionality of the UAV network, its implications can be severe. By intercepting sensitive information, such as navigation commands, system telemetry, or operational parameters, attackers can gain critical insights into the network's architecture and operations. This stolen information can then serve as a foundation for more sophisticated and active attacks, such as spoofing, jamming, or hijacking the UAV system. The passive nature of eavesdropping makes it particularly challenging to detect, emphasizing the importance of implementing robust encryption protocols and secure communication channels to safeguard against such vulnerabilities [54].

B. MAN-IN-THE-MIDDLE ATTACK

The approach man-in-the-middle attack provides an attacker with complete control over the communication channel between a UAV and its command center (illustrated in Figure 4.(a)). Passively, this allows attackers to eavesdrop on information sent from the UAV to the command center and vice versa [64]. Since the attacker also has control over the communication link, they can masquerade as an actual user and communicate with the UAV or command center, potentially appearing as a threat depending on the attacker's intention [16].

C. JAMMING ATTACKS

In jamming attacks they typically target the ability of the communication channel between a UAV and ground control station, effectively rendering the two entities to appear disconnected. In most cases, the UAV's protocols for loss of link would become active, in which the UAV emits messages searching for the ground control station. As long as the jamming attack persists, the UAV is unable to re-establish

communication with the ground station. This prevents the UAV from executing its intended mission, allowing attackers to nefariously exploit the situation [17]. All wireless networks are vulnerable to jamming attacks [35]. This issue stems from the fact that commercial software-defined radios can be easily programmed to function as a jammer, whose effects must be dealt with at the physical network layer. Jamming attacks can affect not only transponder signals, but also any communication network that the UAV relies on such as GPS, satellite communications (SATCOM), or even cellular networks [35]. Jamming attacks are typically classified by how the jamming is performed, while more specific classifications also incorporate the network protocol being used. A generic classification of jamming attacks is presented by [35]:

- **Constant Jamming Attacks** – An attacker broadcasts a powerful jamming signal at all times. This continuously occupies the channel between the transmitter and receiver, preventing any information exchange. The target frequency can be adjusted depending on the attacking process, occupying either the entire channel bandwidth or only a fraction of it.
- **Reactive Jamming Attacks** – An attacker broadcasts a jamming signal only when it detects legitimate packets being transmitted in the channel. This is more energy efficient than a constant jamming attack but requires higher performance equipment to be able to react and send a jamming signal in response to a packet.
- **Deceptive Jamming Attacks** – An attacker sends meaningful signals to the receiver in order to waste the receiver's time and resources, preventing real users from accessing the channel. This is most often performed for Wi-Fi networks.
- **Random and Periodic Jamming Attacks** – An attacker sends jamming signals for random periods of time and then remains idle for the rest of the time. This is more energy efficient than constant jamming but is less effective at disrupting real transmissions on the channel.
- **Frequency Sweeping Jamming Attacks** – An attacker switches between jamming signals at different channel frequencies. This is a workaround used by lower-cost

jammers that cannot attack many channels simultaneously, effectively allowing for lower quality hardware to interfere with more channels.

These generic jamming techniques are applicable to all UAV networks, but the most frequented cases are around GPS jamming. Due to the weak signal strength that GPS signals are received, jamming these signals can be very easily performed by broadcasting noise near the GPS frequency band. A short-range experiment was conducted in [35] and [52], in which an \$8 USD GPS jammer which was purchased from eBay was used to demonstrate the ease and accessibility of GPS jamming hardware.

D. WI-FI ATTACK

A more recent cyberattack against UAVs which makes use of Wi-Fi signals to disrupt the communication channel between a UAV and its ground station, and potentially commandeer the aircraft [65]. Only UAVs that operate on a Wi-Fi signal are susceptible to this type of attack, however given the rising popularity of Wi-Fi for use on drones, this lends itself to becoming a more viable attack strategy (demonstrated in Figure 4.(b)). The attacker first selects a target Wi-Fi network and attempts to gain authorization by launching a de-authentication attack [66]. Once the bypass is successful, the attack is able to hijack the UAV over the Wi-Fi network [16], [17].

E. LARGE-SCALE UAV NETWORKS ATTACKS

Large-scale UAV networks rely on interconnected communication protocols, including ad hoc networks (UAV-MANETs), mesh networks, and cloud-based control systems [67]. While these architectures enhance operational efficiency, they also introduce attack vectors that can compromise the entire network. A single compromised node can escalate into a widespread security breach, enabling various cyber threats. Malware or malicious commands can propagate through wireless communication links, potentially disrupting the entire UAV fleet and causing a DoS attack [40]. Exploiting a single vulnerability in UAV firmware can lead to swarm-wide control loss, facilitating mass hijacking. Additionally, adversaries can intercept and modify command signals, resulting in UAV deviations, mission failures, or complete system takeovers. Furthermore, attacks on the swarm coordination system's consensus mechanism may lead to swarm desynchronization, formation collapse, or collision events, significantly impacting mission success and operational safety [32].

F. FALSE POSITIVE AND FALSE NEGATIVE RATES

The false positive rate (FPR) and false negative rate (FNR) are critical metrics in evaluating the effectiveness of intrusion detection systems (IDS) for various cyberattacks. In Eavesdropping Attacks, where an attacker intercepts confidential data transmissions, IDS may generate false positives due to legitimate users engaging in high-volume data transfers,

while false negatives can occur if the eavesdropping is conducted using passive techniques that do not alter network traffic patterns [54]. Similarly, man-in-the-middle Attacks involve intercepting and modifying communication between two parties. IDS can mistakenly flag legitimate proxy-based communications as attacks (false positives), while sophisticated man-in-the-middle techniques using encryption tunneling may go undetected (false negatives) [16]. For Jamming Attacks, which aim to disrupt wireless communication by overwhelming channels with noise, the false positive rate can rise due to environmental interference (e.g., microwave signals or overlapping Wi-Fi channels), whereas a high false negative rate may result from adaptive jamming techniques that blend into normal traffic patterns [35]. Wi-Fi Attacks, such as deauthentication or rogue access points, can also suffer from high FPR if network administrators frequently reconfigure access points, while FNR can be high when attackers use Media Access Control (MAC) address spoofing to disguise their presence. In Large-scale UAV Network Attacks, including GPS spoofing and signal hijacking, the false positive rate may increase if benign anomalies (e.g., sudden wind changes affecting UAV stability) are misclassified as attacks, while a high false negative rate may arise due to attackers leveraging advanced machine learning-based evasion techniques to bypass detection systems [40].

IV. UAV FUNCTIONALITY ATTACKS

A. UAV CYBERSECURITY STANDARDS

To ensure the safe and secure integration of UAVs into airspace and optimal UAV functionality, regulatory bodies such as the FAA in the U.S. and the EASA have established cybersecurity guidelines [7], [32]. The FAA's Special Condition for small UAVs with Remote Identification mandates that UAVs transmit unique identifiers to enhance accountability and prevent unauthorized drone activities. Additionally, the FAA's UAV Traffic Management framework incorporates cybersecurity requirements to protect UAV communication channels from spoofing, jamming, and unauthorized access [7], [32]. Similarly, EASA's Regulation outline cybersecurity and data protection measures for UAV operations, requiring encryption, secure authentication, and compliance with ISO/IEC 27001 cybersecurity standards to safeguard drone networks from cyber threats [35], [43]. In addition to regulatory frameworks, the Institute of Electrical and Electronics Engineers (IEEE) has developed protocols to standardize UAV cybersecurity practices. The IEEE 802.11s protocol enhances secure UAV-to-UAV communication in mesh networks, reducing vulnerabilities to eavesdropping and interference. Meanwhile, IEEE 1609.2 defines security mechanisms for vehicular and UAV-based communication, ensuring encrypted and authenticated message exchanges in safety-critical applications [39], [40]. Furthermore, the IEEE P1920.2 Standard for UAV Swarm Communication and Security provides guidelines for mitigating cyber risks in large-scale UAV deployments, incorporating cryptographic techniques and blockchain-based identity management [56],

[68]. Aligning with these standards ensures that UAV cybersecurity frameworks remain resilient, interoperable, and compliant with industry best practices.

B. EXPLOITATION OF RECORDED VIDEO

The exploitation of recorded video is a general category pertaining to an attack on the subset of UAVs that use a camera to navigate and avoid collisions. This type of attack firstly requires the attacker to have backdoor access to the UAV's flight controller, which can be achieved through trojans, malware, or other means. This allows the attacker to access the flight controller and systematically replace the real-time camera footage of the drone with a fake view, leading the drone to believe the false video is in fact real. In many cases, the intention of this attack is to trick the drone to land in a different location, allowing attackers to steal the drone [16], [17].

C. DENIAL-OF-SERVICE

A DoS attack is a standard term used to describe intentional communication attacks on a receiver, often by overloading the receiver with information that then forces the system to halt (depicted in Figure 4.(c)). However, in general, denial of service can also include attacks that force the hardware system to work overtime, leading to issues such as memory and Central Processing Unit (CPU) congestion, buffer overflows, and battery exhaustion [16], [69]. Malware, such as hardware trojans or those previously discussed, can lead to denial of service attacks that cause the UAV to malfunction. Such problems, when done at the right time, can lead to the UAV landing, crashing, and shutting down at unsafe times [17]. In the context of UAV Wi-Fi networks, examples of DoS attacks include Transmission Control Protocol/User Datagram Protocol (TCP/UDP) flooding, as well as de-authentication attacks [66]. A TCP/UDP flood involves sending a barrage of TCP or UDP packets to the UAV's corresponding network port, effectively overloading the UAV's processing capability and rendering the UAV unable to respond or perform its tasks. This manner of DoS attack is similar to that applied in regular computer networks [39]. A de-authentication attack involves an attacker impersonating a real user transmitting a de-authentication request, which results in the UAV ceasing communication with the real user and crashing due to loss of controls. The de-authentication request is standardized in the IEEE 802.11 protocol, which is used for Wi-Fi networks, and is meant to allow access points such as the UAV to save on computational resources when a user wishes to end the communication link. However, when this is triggered by an attacker, the user loses control over the UAV, potentially leading to disastrous outcomes [66].

D. GPS SPOOFING

Similar to the other cyberattacks related to message injection, GPS spoofing involves creating a fake GPS signal which is broadcasted at UAVs, causing them to acknowledge a faked

position (see Figure 5). The artificial GPS signals can be created by either ground equipment, or actual satellite that's broadcast at a higher power than GPS satellites. Once the drone is given its falsified position, it can then be hijacked, leading to theft or crashes [17]. Various types of spoofing techniques exist, although the general methodology remains consistent [70]:

- Simple Spoofing – Generating fake Global Navigation Satellite Systems (GNSS) signals for transmission. Can be implemented using low-cost hardware to receive and reproduce GNSS signals, or commercial hardware with greater processing capability. Typically, GNSS is very sensitive to spoofing attacks because of weakness of satellite signals at the earth's surface, in particular these signals are used publicly and not protected.
- Spoofing with high gain antennas – The attacker uses high gain antennas to separate the GNSS signals from noise.
- Intermediate Spoofing – It is an attack through a receiver-spoofers and the attacker generates false GNSS signals while simultaneously attempting to attack the target receiver through code phase alignment between the fake and real signals. The receiver will track the satellite signals to synchronize as accurate as possible using the satellite time and estimate the Doppler frequencies and code phases of each satellite signal tracked by the victim receiver.
- Spoofing with multiple transmitting antennas – An advanced technique used against receivers that have multiple antennas. The attacker uses multiple antennas to attack each of the receiver's antennas. Every transmitting antenna of the attacker is directly related to corresponding receiving antenna on the victim side.
- Sophisticated spoofing – Performed by groups of attackers, who coordinate and synchronize their attacks on the receiver GPS system. They are able to attack the victim's receiver in an efficient as well as organized manner. They can have 3D position data of the victim's antenna and in turn can overcome complex countermeasures, for instance angle of arrival estimation.

This is much more effective at overcoming complex anti-spoofing countermeasures, such as techniques that employ estimation of angle of arrival. The process of creating a portable GPS spoofing system with low-cost SDR equipment and publicly available code frameworks is presented in [70] and [71]. The investigation concluded that, due to the lack of protection for GPS receivers against spoofing, it was possible to spoof a receiver's positions quite easily. Note however that this is not legal, and should not be attempted without authorization.

V. UAV ELECTRONIC WARFARE FOR MILITARY AND CIVILIAN PURPOSES

In the above sections the discussion has focused on the civilian side of UAV cyberwarfare, however it should be noted that those same tactics are applicable to military UAVs as

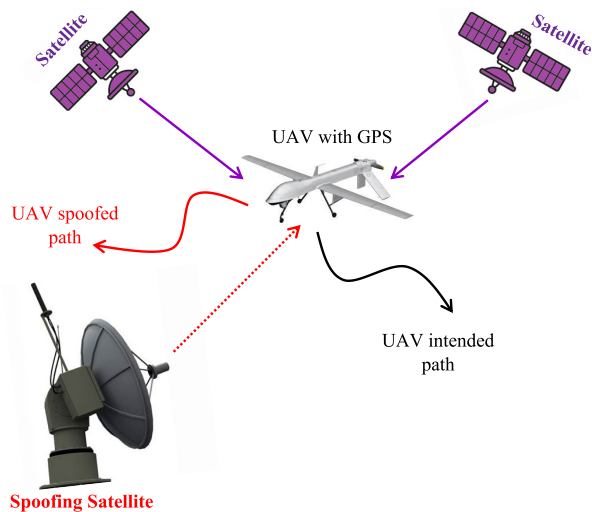


FIGURE 5. GPS spoofing attack.

well. This section starts first by discussing support measures, countermeasures, and counter-countermeasures. Next, some notable UAV cyberattacks occurrences are presented.

A. SUPPORT, COUNTERMEASURES, AND COUNTER-COUNTERMEASURES

For military drones, UAV electronic warfare includes not only the previously listed cyberattacks, but also intelligence and counterintelligence operations employing the use of UAVs to carry out a specific signaling mission. Intelligence related missions typically involve setting up UAVs to spy on enemies for indefinite periods of time, acquiring information regarding enemy movements, routines, and patterns such that future attack can be planned with a high degree of situational awareness. Part of this intelligence acquisition relates to safeguarding the electromagnetic spectrum for friendly use, and disallowing enemies from having control over the spectrum [52]. Details of UAV usage in the military for electronic warfare are typically not published publicly for obvious reasons. However, given that the advantages of using an unmanned aerial vehicle for militaristic are plentiful, it should come as no surprise that its applications in this realm have not been fully explored. Classical electronic warfare was divided into categories of EM Support Measures, EM Countermeasures, and EM Counter-countermeasures [72]. The USA and NATO have updated these categories to reflect modern advancements in electronic warfare [73]:

- ES – Electronic Warfare Support (Formerly EM Support Measures): It deals with collecting enemy signals, communications or otherwise, for immediate action (jamming, location determination, etc). Its main objective is to intercept, locate, identify and/or localize source of unintentional radiated EMS for the sake of immediate threat.
- EA – Electronic Attack (Formerly EM Countermeasures): EA also includes other weaponry such as

anti-radiation weapons and directed-energy weapons. EA main goal is to attack personnel, equipment, or facilities, with the focus on degrading, neutralising, or destroying the enemy capability.

- EP – Electronic Protection (Formerly EM Counter-countermeasures): Alongside these categories is Signal Intelligence (SIGINT) which involves the reception of enemy EM transmissions. SIGINT is composed of Communications Intelligence (COMINT – receiving enemy communications signals to extract intelligence) and Electronic Intelligence (ELINT – using enemy's communications patterns to develop countermeasures) [72].

B. NOTABLE OCCURRENCES OF UAV CYBERATTACKS

Let us explore some of notable occurrences of UAV cyberattacks over the last three decades. These notable instances of UAV cyberattacks were reported from civilian and military sectors and they are listed below:

- 2013 – Hak5, a popular cyber and informational security company, demonstrated numerous vulnerabilities in UAV systems, ranging from using drones as Wi-Fi eavesdroppers to a DoS attack that forced the drone to drop out of the sky. Among these experiments was one which allowed drones infected by malware to seek out and infect other drones as well, providing total takeover of the control system [52].
- 2012 – A GPS jamming attack caused an S-100 Camcopter UAV to crash into the ground control van, killing a one engineer and injuring two remote pilots during testing. This occurred in Incheon, South Korea, from an unknown attacker [74].
- 2011 – the U.S. military's RQ-170 Sentinel UAV was hijacked in Iran, leading to the drone's capture and breach of informational security. Although the method of hijacking was never confirmed, researchers afterwards proved that it was possible to hijack drones through GPS spoofing [75]. It is suspected that a combination of GPS jamming and spoofing led to the UAV being captured.
- 2011 – A Nevada ground control station manning a UAV network was infected by a keylogger. No known classified information was lost or transmitted [76].
- 2009 – Iraqi forces intercept a video stream being transmitted from a UAV to its ground control station. The unsecured communication link was intercepted using SkyGrabber, an off-the-shelf product for use with satellite feeds [64].
- 1996 – An Israeli UAV's video feed was allegedly intercepted by Hezbollah militants, assisting the forces in ambushing and killing Israeli commandos [77].

In a summary of detailing instances of UAV cyberattacks, it was found that GPS spoofing attacks were the most common method, with GPS jamming coming in second. Less common attacks included de-authentication attacks, zero-day

vulnerabilities, exploitation of recorded video, interception of data feeds, and virus attacks.

C. ELECTRONIC WARFARE IMPACT ON CIVILIAN UAVS

Civilian UAV applications, such as disaster response, environmental monitoring, and precision agriculture, have witnessed significant growth in recent years due to advancements in UAV technology and artificial intelligence [32]. Ensuring cybersecurity in these applications is crucial to maintaining operational safety, protecting sensitive data, and ensuring sustainable deployment. UAVs play a critical role in disaster response by providing real-time surveillance, assessing damage, and locating survivors. However, a lack of robust cybersecurity measures can lead to system intrusions, resulting in misinformation or loss of control. Cyberattacks on UAVs used in disaster zones can compromise mission-critical data and endanger lives. Implementing strong encryption protocols and secure communication channels ensures data integrity and reliable decision-making. Research emphasizes the role of cybersecurity in UAV systems to safeguard data transmission and maintain operational trustworthiness [7]. Environmental Monitoring UAVs are increasingly used for environmental monitoring tasks, such as tracking deforestation, wildlife, and pollution levels. These systems collect sensitive data that, if intercepted, could be misused. Cybersecurity frameworks protect this data, ensuring confidentiality and preventing unauthorized access. Advances in cryptographic techniques and secure data storage methods can mitigate the risk of data breaches. Studies show that UAV data security is essential for maintaining the credibility of environmental monitoring programs [78]. In precision agriculture, UAVs collect data on soil conditions, crop health, and resource usage, enabling data-driven farming decisions. Cyberattacks on UAV networks could lead to altered data or service disruptions, potentially causing significant economic losses. Implementing cybersecurity measures, such as multi-factor authentication and intrusion detection systems, enhances the reliability and safety of agricultural UAV operations. Research highlights that secure UAV systems improve overall sustainability by ensuring data accuracy and protecting against malicious interference [79]. Cybersecurity Enhancing Sustainability and Safety Overall, cybersecurity measures not only prevent unauthorized access and data breaches but also ensure the long-term viability of UAV operations by maintaining system integrity. This directly contributes to operational safety, as secure UAVs are less prone to hijacking or failure due to cyber attacks. Moreover, ensuring data integrity and availability supports sustainable practices, particularly in precision agriculture and environmental monitoring, where data accuracy is paramount for decision-making.

UAVs are widely deployed for air quality monitoring, especially in regions prone to industrial pollution or wildfires. These drones collect real-time data on pollutants like CO_2 and particulate matter. However, their reliance on wireless

communication makes them susceptible to electronic warfare attacks, such as GPS jamming or spoofing. An attack compromising their navigation systems could not only disrupt data accuracy but also increase energy consumption through re-routing or signal loss recovery processes. During the 2020 wildfires in California, UAVs were instrumental in mapping air quality and guiding firefighting strategies. An incident reported in 2017 demonstrated how GPS interference during a test flight in Nevada affected UAV stability and fuel efficiency [80]. This shows the dual need to secure these systems against electronic threats to ensure reliable environmental data collection and minimize unintended environmental impacts due to system failures. Electronic warfare and cybersecurity operations often involve extensive use of data centers, which have substantial energy demands. Efficient cybersecurity practices, such as optimized encryption protocols, can reduce the computational burden and thereby lower energy consumption. For instance, adopting lightweight cryptographic methods can cut power usage by 20% compared to traditional encryption [81].

VI. COUNTERMEASURES AND DEFENSIVE-AIDS

The idea behind countermeasures and defensive-aids is to protect the data handled by UAVs in terms of confidentiality, integrity, and authenticity as well as to guarantee the availability of service [1], [21], [22], [26], [27], [28], [29] in both civilian and military applications. Considering the variety of UAV cyberattacks discussed in Section II-V, naturally over time a wide spectrum of countermeasures to these attacks were created to cater to vulnerabilities found on hardware, software, and network layers. Because the same countermeasure method can work against multiple types of attacks, these methods are discussed in broad categories: prevention, detection, communication traffic, flight behavior, mitigation, inertial navigation (see Figure 1(c)). Each category includes specific countermeasures which are discussed in more detail with respect to the aforementioned attacks.

A. PREVENTION

Prevention methods focus on making it difficult to attack the UAV in the first place. The three main subgroups of preventative methods include “Access Control”, “Information Protection” and “Component Selection”.

1) ACCESS CONTROL

The essence of access control is to ensure that the UAV can be contacted only by authorized personnel or software. This can be done with password-based node authentication schemes. One example of realization of this method, in case if Wi-Fi is used as a communication protocol, is to establish MAC address of a device that is trying to access the UAV as a password. MAC is a unique hardware identifier that is attributed to each electronic device on a network, therefore if specific MACs are predefined for access prior to mission, any other authenticating messages will be rejected by UAV [82].

This type of countermeasure can prevent GPS spoofing and Wi-Fi attacks [21].

2) INFORMATION PROTECTION

As discussed prior, the messages involved in UAV links can be intercepted, eliminated or infused. Several methods can be employed to protect the confidentiality and integrity of the messages sent and received by UAVs. One of the most widely discussed method is usage of cryptography, particularly because most UAVs operate over insecure data links such as GPS and Wi-Fi. There are two main types of encryption schemes: asymmetric and symmetric. Asymmetric schemes make use of a public key (used for encrypting the message) and the private key (used for decrypting the message) whereas in symmetric schemes the same key is used for encrypting and decrypting the message [22]. The asymmetric encryption is more computationally demanding and can be challenging to implement on UAVs with limited resources [26], hence the symmetric schemes might be preferred. Another way to interfere with message integrity is jamming. In order to protect the link from jamming spread spectrum techniques are utilized, such as Frequency Hopping (FH), Frequency Hopping Spread Spectrum (FHSS), and Direct Sequence Spread Spectrum (DSSS). Spread spectrum techniques involve using random sequences to spread the message over a wide band of frequencies which makes jamming ineffective since it blocks only one specific frequency at a time. The sequence is shared between the sender and receiver which then allows for “decoding” the message [83].

3) COMPONENT SELECTION

The component selection strategy relies on designing or sourcing components that do not have exploitable vulnerabilities. This can be achieved on software, firmware/sensors levels, however it could be quite challenging to achieve on all levels as it could potentially drive the cost and complexity up. Instead, some anti-tampering technologies can be employed on board of the UAVs to prevent entry points for potential attacks [26].

B. DETECTION AND RADIO SIGNAL CHARACTERISTICS

Once the prevention methods fail, it is crucial for UAV to be able to detect and identify the type of attack that it is experiencing. The largest subgroup in detection methods is related to “presence of anomalies”.

1) PRESENCE OF ANOMALY

Cyberattacks often cause abnormal patterns, a variety that can be observed in radio signals, communication traffic, flight behaviour or the surrounding environment. These anomalies can help detecting an attack on a UAV of interest. Cyberattacks often target the communication link between UAVs and their control stations. This can cause anomalies in the radio signals. Unusual changes in signal strength

might indicate jamming or spoofing attacks. The appearance of unexpected frequencies or distortions in the spectrum can suggest intentional interference. Prolonged delays or missing packets in telemetry data transmission may indicate eavesdropping or DoS attacks. The data exchanged between a UAV and its ground station or other UAVs is a prime target for attackers. A surge or drop in data transmission could signify an attempted intrusion or data exfiltration. Deviation from standard communication protocols may indicate malicious attempts to compromise the system. Access from unauthorized or suspicious IP addresses can suggest a potential breach.

2) RADIO SIGNAL CHARACTERISTICS

GPS spoofing causes sudden changes in received signal such as signal strength, noise level, signal phase-delay, etc. One of the methods to detect GPS spoofing attack is to set a threshold value for signal strength that will determine if the incoming signal is too strong and hence is a spoofing attempt [26]. This simple method is also enhanced by [67] through dynamic changing of threshold value using machine learning algorithm (e.g., Support Vector Machine (SVM)) which counters attacker’s attempt to change the transmitted signal strength and bypass the threshold.

C. COMMUNICATION TRAFFIC

Message elimination and infusion may cause anomalies in the communication traffic. For example, the data packet delivery ratio can drop significantly because of message elimination or, on the contrary, as a result of message injection attack there can be a surge of incoming messages [26]. One of the methods to detect attacks of this type is to have a mechanism which allows the UAVs in the network to compare their traffic parameters with neighboring UAVs. This method is also effective against spoofing attempts [27], [32]. Additionally, machine learning techniques are used to detect general anomalies in network traffic by taking in parameters such as packet size, flow duration, number of packets. Study in [28] compares the effectiveness of various machine learning algorithms and finds that decision tree algorithm is the most accurate compared to other algorithms.

D. FLIGHT BEHAVIOUR AND UAV ENVIRONMENT

GPS spoofing often causes changes in UAVs trajectory or other flight parameters as its main goal is to lead the UAV away from its intended course. If a UAV has a very predictable and repeatable path, then an algorithm based on flight statistics can be employed to detect any anomalies in flight profile, attitude or thrust [84]. This method is quite limited as it would not perform well for UAVs that do not operate over repeated path. In addition to changes in flight parameters, GPS spoofing can also cause errors in calculated position coordinates. Such errors can be detected by using UAV model estimator in conjunction with position sensors on board

of UAV [85]. Once the error reaches certain threshold the system issues an alarm signal. To improve the performance a Kalman filter can be added to the estimator as proposed by [86] in order to better deal with uncertainties in UAV movements. Finally, the last reference that the UAV can use to detect positional anomalies is the surrounding environment. One of the easier methods to detect spoofing attempts is to simply compare the coordinates and imagery outputs from the UAV. If there is an inconsistency between the two, it can be concluded that the UAV might be either under spoofing or video replay attack [82]. Another interesting method to deal with video replay attacks is to employ models that determine the solar shadow position of the UAV according to UAV's location, the sun position and the current time [29]. Inconsistency in expected solar shadows indicate that the video output has been hijacked by the attacker.

E. MITIGATION

As the attacks are identified the UAV can move on to mitigation strategies to overcome the attack. The main subgroups in the mitigation methods are Neutralize/Avoid, Redundancy and Fail-Safe Protocols.

1) NEUTRALIZE/AVOID THE ATTACKER

Neutralizing the attacker can be particularly effective when the message spoofing or eavesdropping attack has been identified. The UAV can launch a counter-jamming attack which would overwhelm the attacker's receiver and discontinue the attack. The main limitation is potential consequential jamming of friendly UAVs nearby [21], [61], [82]. The study in [63] proposed cooperative jamming where one friendly UAV is deployed as a jammer as close as possible to the eavesdropper. This method, however, relies on the fact that the eavesdropper's location is known or easy to determine, which might not be the case. Avoiding the attacker can be realized in different ways. It can be as simple as escaping the adversary's RF coverage range in case of a jamming, eavesdropping or spoofing attacks. Another avoiding strategy is to divert the attack. It can be done by creating a device that emits signals that closely resemble the signal qualities of UAV but transmitted at a higher power. The concept is known as honeypot and was proposed in [87].

2) PROVIDE REDUNDANCY

Redundancy can be ensured on various levels to mitigate attacks. For instance, in case of GPS jamming the UAV can switch to a different GNSS constellation such as Galileo or GLONASS if it is provided with antennas suitable for different GNSS signals. This method is deemed quite effective as it is not easy to jam all GNSS signals simultaneously [88]. Redundancy can also be provided on sensor level to prevent dispatch system attacks. For instance, an UAV can have both an optical and a MEMS gyroscope to ensure the attitude input in case one of the gyroscopes is compromised.

3) FAIL-SAFE PROTOCOLS

Fail-safe protocols are critical safety mechanisms designed to protect UAVs and their payloads when communication with the ground station is lost, or other system anomalies occur. In case UAV has lost the connection with the ground station and is not able to achieve recovery, it can resort to predetermined protocols such as "Return to Homebase" or "Self-destruct", to prevent the enemy from capturing the UAV [89]. These protocols act as the last line of defense to prevent unauthorized access, capture, or exploitation of the UAV and its systems.

F. STRIDE: A CYBERSECURITY THREAT MODELING FRAMEWORK

Spoofing, Tampering, Repudiation, Information Disclosure, DoS, and Elevation of Privilege (STRIDE) is a threat modeling framework developed by Microsoft to systematically identify and mitigate security threats in software, hardware, and cyber-physical systems including drones [90], [91], [92]. STRIDE is widely utilized in cybersecurity to analyze potential vulnerabilities and design appropriate countermeasures [92]. The framework categorizes six primary security threats: spoofing, tampering, repudiation, information disclosure, DoS, and elevation of privilege. To mitigate these threats, various security measures are implemented. Spoofing is addressed through multi-factor authentication, digital signatures, and encryption. Tampering is countered using integrity checks, digital signatures, and cryptographic hashing. Repudiation is managed through secure logging, audit trails, and digital signatures. Information disclosure is mitigated by implementing data encryption and access control policies. DoS attacks are prevented through redundancy, network filtering, and anti-jamming measures. Elevation of privilege is mitigated by enforcing role-based access control, adhering to the principle of least privilege, and patching vulnerabilities. STRIDE provides a structured approach to identifying potential cybersecurity threats, helps to design security countermeasures before a system is deployed, and applicable to software security and drones cyber-physical systems. Table 3 presents security aspect of transponder-based cybersecurity and STRIDE framework [90], [91], [92].

G. FULL AUTONOMY AND INERTIAL NAVIGATION

1) AVOID WIRELESS COMMUNICATION

UAVs are equipped with exteroceptive and proprioceptive sensing units to facilitate trajectory planning, navigation, and tracking control. UAVs are typically attached with a variety of sensors, such as, GPS, IMU, laser lightning and Light Detection and Ranging (LiDAR), vision units (stereo, monocular, or RGB-D cameras), Ultra-wideband (UWB) tags and anchors, and ultrasonic sensors [4], [93], [94]. GPS sensor enables satellite-based navigation providing a vehicle navigating in 3D space with its position, heading/direction, speed, and time information given the availability of at least

TABLE 3. Transponder-based cybersecurity vs. STRIDE framework.

Security Aspect	Transponder-Based Cybersecurity	STRIDE Framework
Focus	Authentication, identification, and UAV tracking.	Threat modeling identifies six security threats.
Threat Coverage	Addresses spoofing, tampering, and repudiation threats.	Covers spoofing, tampering, repudiation, DoS, and others.
Spoofing Mitigation	Uses encrypted identifiers and authentication to prevent UAV spoofing.	Identifies spoofing and suggests countermeasures, e.g., multi-factor authentication.
Tampering Resistance	Implement data integrity and cryptographic measures to prevent tampering.	Uses cryptographic mechanisms.
Repudiation Prevention	Provide verifiable identity logs for auditing, reducing the risk of denial by malicious actors.	Suggests maintaining secure logging and digital signatures for non-repudiation.
Information Disclosure Protection	Designed for identification rather than secrecy.	Highlights the need for encryption and prevent data leaks.
DoS Mitigation	Vulnerable to jamming attacks and requires redundancy or anti-jamming measures.	DoS-resistant architectures, e.g., rate-limiting, redundancy, and anomaly detection.
Elevation of Privilege Protection	Do not directly prevent privilege escalation attacks but enforce access control policies.	Suggests strict access control and privilege enforcement.

four different satellites (for 3D navigation) or at least three different satellites (for 2D navigation) [4], [93]. However, GPS tends to be unreliable in certain scenarios, for instance, GPS is unavailable indoor and it is subject to obstructions, multipath, fading, and signal-denial [4], [93]. GPS also has additional challenges such as receiver clock error, satellite clock error, Tropospheric delay, Ionospheric delay, receiver noise, satellite orbital (ephemeris) errors, and errors due to satellite geometry. IMU is used for orientation determination of the vehicle. IMU solely can be used for dead reckoning (determination of orientation, position, and linear velocity) to avoid wireless communication, however it is only reliable for short-path navigation, however unreliable for long-path navigation due to error drift and accumulation [95]. UWB sensors have similar concept to satellite-based positioning. UWB tag(s) can provide the UAV with its position in 3D space via predefined fixed or moving anchors using wireless communication [4]. The main limitation of UWB units is their susceptibility to measurement noise [4]. Onboard UAV vision units offer position localization without wireless communication. Vision units work by instantaneous feature points tracking of two different frames (captured by stereo camera) or via two consecutive frames (obtained by monocular camera) to determine vehicle's position with respect to a known earth-frame [93], [95], [96]. Vision units could be challenged in the event of low-texture environments or in

case of high altitude flights which may result in degradation in positioning accuracy. LiDAR is a distance measurement sensor which has a similar concept to radar such that instead of employing radio waves. LiDAR utilizes light that hits the target and reflects and helps in distance determination. LiDAR can face challenges through the necessity of large amount of data to provide high accuracy. Also, LiDAR faces difficulties in unstructured areas (e.g., machinery zone and shelving). In addition, LiDAR is an expensive unit when compared to other navigation sensors which makes it unfit for low-cost UAVs [32].

2) MULTI-SENSOR FUSION

The sensor fusion helps to fuse different sensors to improve full autonomy and enhance navigation accuracy, and in turn it helps to accomplish what a single sensor cannot do. In navigation, multi-sensor fusion is coupled with estimator or filter design to reject sensor uncertainties and provide good estimation accuracy including the hidden states such as the vehicle's linear velocity [95]. Examples of multi-sensor fusion applied in UAV avionics applications for navigation purposes include GPS-IMU [97], vision-based navigation (vision unit + IMU) [1], [33], [93], [95], UWB-IMU [4], LiDAR-based IMU [98], and others. As mentioned previously combination of different types of multi-sensor fusion are used to improve estimation accuracy and reliability. Note that each sensor has its own frequency of data measurement collection (typical low-cost IMU has a rate of 200 Hz and low-cost stereo camera can provide photographs at a rate of 20 Hz) [93], [95]. Thus, multi-sensor fusion and filter design should account for data collection variation frequency of different sensors and signal processing.

H. CASE STUDIES

Cyberattacks, such as GPS spoofing or jamming, could misdirect UAVs or disrupt communication, delaying critical assistance. A case study demonstrated that implementing encryption protocols and anti-jamming technologies significantly enhances UAV resilience during disaster management missions, ensuring continuous operation and accurate data transmission [35]. In terms of logistics and delivery services, several companies are testing UAV-based delivery services, where secure data transmission is crucial for package tracking and delivery confirmation [99]. Man-in-the-middle (MITM) attacks could intercept or manipulate this data, leading to package theft or misdelivery. Robust encryption and authentication mechanisms are vital to counter such threats. UAV communication frameworks in a case study, showing how Transport Layer Security (TLS) protocols prevent unauthorized access and data tampering in drone delivery networks, thereby safeguarding logistical operations [100]. With regard to environmental monitoring, UAVs monitor environmental parameters such as air quality, wildlife movements, and deforestation. These missions depend on the integrity and accuracy of collected data. Cyberattacks

TABLE 4. Summary of cyberattack threat focus and difficulty.

Attack	Type	Threat Target	Level
Dispatch System Attack	Active	Confidentiality, Integrity, Availability, Privacy	High – Requires installation of malware or trojans designed for the UAV.
ADS-B Attack	Active	Confidentiality, Integrity, Privacy	Medium – Used commercially but requires knowledge of ADS-B transmission protocols.
TCAS Induced Collision	Active	Integrity, Availability	High – Requires another attack (malware or misguidance) to enable collision to occur.
TCAS Attack	Active	Availability	High – Attacker must be familiar with the equipment (transmit TCAS messages) and protocols.
Eavesdropping	Passive	Confidentiality, Privacy	Low – Can be done using receiver such as SDR. Easier if communication channels are not encrypted.
Man-in-the-Middle Attack	Passive	Confidentiality, Integrity, Availability, Privacy	High – Attacker must insert a device between UAV and ground stations.
Jamming	Active	Availability	Low-Med – Can be done by transmitting high powered signals at set frequencies. More difficult if sophisticated methods are used.
Wi-Fi Attack	Active	Confidentiality, Availability, Privacy	Med – Requires knowledge of the Wi-Fi protocol and a transmitter.
Exploitation of Recorded Video	Active	Confidentiality, Integrity, Privacy	High – Requires the attacker to install malware on the UAV and relay false videos to the kernel.
Denial of Service	Active	Availability	Med – Attacker can spam UAVs with useless packets. Requires a transmitter and other hardware.
GPS Spoofing	Active	Integrity, Availability	Low-Med – Attacker must transmit GPS signals. Such devices are available in commercial markets.

targeting data integrity could manipulate environmental data, leading to incorrect assessments or policy decisions. In a real-world scenario, the use of blockchain technology to ensure data integrity in environmental monitoring UAV networks [101]. Blockchain's decentralized nature helps prevent unauthorized data alterations, ensuring that collected environmental information remains trustworthy and verifiable.

VII. COMPARISON OF UAV CYBERATTACKS AND COUNTERMEASURES

In this section, the cyberattacks and their countermeasures are compared in terms of their effectiveness and ease of implementation on UAVs.

A. COMPARISON BETWEEN DIFFERENT CYBERATTACKING STRATEGIES

The various cyberattacking strategies presented in Section II-V were detailed in a high-level overview of the cyberattacking methodologies most applicable to UAVs. This section elaborates more on those cyberattacks by providing a comparison of their attacking strategies, threat focus, and difficulty of execution. It should be noted that these cyberattacks are not specific towards UAV or transponders, however the discussion will still revolve around the usage of attacks on UAVs. Table 4 shows a brief comparison of the various cyberattacks that were mentioned in previous sections [10], [34], [43], [102], [103], [104], [105], [106], [107]. These attacks comprise the main category of attacks,

meaning more granular subcategories would be able to identify more attacks than those listed here.

In the context of UAVs, most cyberattacks focus on actively violating the confidentiality of the UAV or impairing its availability. In most cases, this is done through wireless means such as through radio waves, such as for ADS-B attacks, TCAS induced collisions, jamming, Wi-Fi attacks, and denial of service. These wireless attacks are more prevalent since modern SDR are capable of transmitting or receiving at a wide range of frequencies and can be programmed to transmit complex protocols by malicious users. For instance, some SDRs [108] available online even contain pre-programmed transmission and receive protocols for ADS-B, ACARS, etc. Such devices are typically not barred from country imports due to the very general use of SDRs, however malicious users are capable of using these devices for nefarious purposes.

B. COMPARISON BETWEEN COUNTERMEASURE STRATEGIES

This subsection provides a comparison summary of all countermeasure strategies discussed in Section II in terms of attacks that they are effective against as well as their limitations. Table 5 provides summary of all preventive countermeasures, where it can be seen that the most versatile preventative methods are node authentication and encryption. In context of UAVs, the only concern is the limited computational power which can be a significant limitation for employing cryptographic methods on board of smaller UAVs.

TABLE 5. Prevention methods comparison.

Prevention	Attack Types	Limitation
Node Authentication	Wi-Fi attacks, GPS spoofing, Dispatch system attack	Easy to implement solution, however, effectiveness depends on password complexity
FHSS and DSSS	Jamming	Effectiveness diminishes as number of UAVs increases
Cryptographic Encryption	ADS-B Attack, Eavesdropping, Man-in-the-Middle	Computationally demanding (limited on board computational power)
Tamper-proof components	Dispatch System Attack	Additional cost, difficulty sourcing completely tamper-proof devices

TABLE 6. Detection and mitigation methods comparison.

Prevention comparison		
Prevention method	Attacks	Limitation
Threshold method	GPS spoofing	Easy to bypass by an adversary
Machine learning	ADS-B attack, Message deletion, GPS spoofing, DoS attack	Can be computationally demanding
Flight statistics	GPS spoofing	Relies on predictable UAV trajectory
Reference based	GPS spoofing	Can be computationally demanding
Image comparison	Message injection, GPS spoofing	High dependence on communication links (UAV-to-Ground or UAV-to-UAV)
Mitigation comparison		
Mitigation method	Attacks	Limitation
Jamming counterattack	Eavesdropping, message spoofing/injection	Jamming could potentially interfere with the UAV itself
Escaping the coverage range of an adversary	Jamming, spoofing, message injection	Uses limited power resources
Adding multiple antennas for redundancy	Jamming, sensor (dispatch system) attack	Physical limitations of UAV (available space, power etc.)
Implementation of fail-safe protocol	Jamming, ADS-B attack, Dispatch system attack	Last resort means (provides no link or data recovery)

The upper part of Table 6 presents summary of detection methods, which shows that machine learning algorithms proved to be quite effective for detecting multiple types of attacks by using RF signal and communication traffic data. Just as in case with encryption the only limitation for implementation of machine learning algorithms on UAV is their limited on-board computational power. Finally, the lower part Table 6 provides with comparison of mitigation methods. All of these methods are effective against multiple

types of attacks; therefore, it is not possible to provide a simple recommendation on which mitigation method is best.

Node authentication and encryption are highly versatile, as they address a wide range of cyber threats, including eavesdropping, spoofing, and data tampering. Smaller UAVs often lack the processing capacity to execute complex cryptographic protocols in real-time. Implementing robust encryption may introduce delays in time-sensitive applications such as navigation or real-time surveillance. Machine learning-based methods stand out for their ability to detect a wide array of attack vectors, including jamming, spoofing, and unauthorized access. Training and deploying machine learning models require significant computational resources, which are often unavailable on smaller UAVs. Machine learning systems can misidentify benign anomalies as threats, potentially disrupting operations unnecessarily. All mitigation methods are effective against multiple attack types; however, their suitability depends on specific mission requirements and threat models. Advanced reactive systems may demand significant computational and energy resources. Fail-safe protocols, while effective, may disrupt mission objectives or result in UAV loss under certain conditions. In general, despite providing suggestions on more effective countermeasures, the final choice of methods largely depends on the UAV mission and the attacks that can pertain to that mission [32]. For instance, in case of stationary UAV swarms that are used for communication networks, perhaps securing the traffic with machine learning techniques makes a lot more sense than for a singular drone that performs delivery and would benefit more from GNSS connectivity redundancy to ensure reliable navigation.

VIII. FUTURE TRENDS

As drones become more prevalent in our daily lives so is the risk and complexity of cyberattacks. The current trends in the cybersecurity have been discussed in previous sections, however as it was seen even the most novel methods run into significant limitations associated with limited size or resources available to UAVs. Hence this section briefly introduces the potential future research trends that pertain to the UAV cybersecurity.

A. COMPUTATIONALLY EFFICIENT MACHINE LEARNING ALGORITHMS

As was shown in Table 6, the limited computational resources available to UAVs are a significant limitation when it comes to employing computationally heavy algorithms (i.e., machine learning or artificial intelligence). This can be remedied by outsourcing training phase to the ground which would diminish the algorithm to a simple mapping function [109]. Such approach is limited since no new learning can be introduced online, therefore developing computationally efficient machine learning algorithms that can be implemented on board of a drone is an important research vector for UAV cybersecurity.

B. LACK OF DATASETS FOR FEDERATED LEARNING MODELS

While traditional machine learning methods use a centralized data source to train the model, Federated Learning (FL) method uses multiple entities that train a model. FL methods are actively researched in context of UAVs since they allow for stronger data privacy protection as well as network scalability (increase in UAV numbers). The biggest issue that prevents FL models from being immediately implemented on UAVs is the lack of drone datasets that can be used for training these models, i.e., network traffic datasets, and malware datasets [110]. Therefore, the future research will very likely focus on generating more datasets that can be used for effective model training [111].

C. AI AND ADVERSARIAL MACHINE LEARNING IN UAV SECURITY

AI-generated cyber threats are becoming a significant challenge in UAV security [30], [31], [32]. Attackers are leveraging AI to automate and enhance cyberattacks, making them more adaptive, stealthy, and difficult to detect. Adversarial Machine Learning (AML) techniques are being integrated into UAV security models. AML-based security frameworks can detect and neutralize adversarial AI threats by training UAV-based IDS on adversarially generated attack scenarios. For instance, defensive AI models can employ adversarial training to improve robustness against evasion attacks, where attackers modify malicious inputs to deceive detection algorithms. Furthermore, FL and decentralized AI approaches enable UAVs to collaboratively update their security models without exposing raw data, enhancing both privacy and adaptability. By incorporating generative adversarial networks (GANs) and explainable ML approaches for anomaly detection and reinforcement learning-based defense mechanisms, UAV security systems can proactively adapt to evolving AI-generated threats, ensuring resilient and autonomous threat mitigation in dynamic environments.

D. FEASIBLE CRYPTOGRAPHIC METHODS

Cryptographic methods, although are quite effective for preventing cyberattacks, run in the same limitation as machine learning techniques, which is restricted by computational power limitations. As was seen in Section VI-A out of two major encryption schemes the symmetric schemes are more preferred to asymmetric ones since they are less computationally demanding. However, symmetric schemes rely on reliable key distribution method which in itself can require a lot of computational resources. The work in [62] suggest distributing keys to UAV when they are on the ground and using that one distributed key over the duration of flight. A distribution like that would require a physically secure channel which can be realized via Quantum Key Distribution (QKD). Study in [62] describes

implementation of BB84 QKD for UAV applications. Despite QKD being very safe, it is still prone to man-in-the-middle attacks which is something to be addressed in the future research.

E. SECURING DATA AGGREGATION PROCESS

As the networks of UAVs continue to expand, so does the volume of data exchanged between UAVs and their respective ground stations. To optimize both the energy consumption of UAVs and the associated communication costs, data aggregation emerges as a critical process. This involves collecting and consolidating data from multiple sources to minimize redundant transmissions and maximize efficiency. However, ensuring the security of this aggregated data is paramount, given the risk of interception or tampering during transmission. Secure data aggregation protocols not only reduce overheads but also safeguard the integrity and confidentiality of the information exchanged. Although advancements are being made in this field, it remains a vibrant area for future research, as evolving technologies and threats necessitate continuous improvement and innovation in secure and efficient data aggregation methods [41].

F. BLOCKCHAIN-BASED UAV AUTHENTICATION

Blockchain technology has emerged as a robust solution for securing UAV networks by enabling decentralized authentication and secure communication frameworks [101]. Traditional authentication mechanisms rely on centralized authorities, which introduce vulnerabilities such as single points of failure and susceptibility to cyberattacks. By leveraging blockchain, UAV networks can achieve tamper-resistant, transparent, and trustless authentication, where each UAV is registered as a node with cryptographic credentials stored on an immutable ledger [56]. Smart contracts can be used to automate authentication processes, ensuring only legitimate UAVs can access network resources while preventing identity spoofing and unauthorized control [68]. This approach significantly enhances security in large-scale UAV swarms and heterogeneous drone networks. In addition to authentication, blockchain facilitates secure and resilient communication between UAVs and ground control stations. By integrating consensus mechanisms such as proof-of-work, proof-of-stake, or lightweight alternatives such as delegated-proof-of-stake, UAV networks can verify message integrity and prevent data tampering [68]. Furthermore, end-to-end encryption and decentralized key management ensure that communication remains confidential, mitigating risks from eavesdropping, man-in-the-middle attacks, and jamming attempts. The distributed nature of blockchain also enhances network resilience against DoS attacks, ensuring continued UAV operations even in adversarial environments. By combining blockchain with edge computing and AI-driven anomaly detection, UAV networks can achieve real-time, adaptive security against emerging cyber threats.

G. CRYPTOGRAPHIC PROTECTIONS AND FLIGHT ENDURANCE

Cryptographic protections in UAVs enhance security by encrypting communication, authenticating signals, and preventing cyber threats such as spoofing and jamming. However, these protections impose computational overhead, increasing the power consumption of onboard processors. This additional energy demand can strain the UAV's limited battery capacity, reducing overall flight endurance. The impact varies based on encryption algorithms, hardware efficiency, and system optimization. Lightweight cryptographic schemes and dedicated security hardware can help mitigate energy drain, balancing security with operational efficiency. Thus, while cryptographic protections are essential for secure UAV operations, they must be carefully designed to minimize their impact on battery endurance.

IX. CONCLUSION

Cyberattacks applicable to Unmanned Aerial Vehicles (UAVs) were comprehensively reviewed and discussed, with their impacts and methodologies summarized at a high level. The majority of cyberattacks targeting UAV networks attack the system through wireless communications, with the more prevalent attacks being those non-specific to any transponder. Examples of these include jamming, spoofing, and eavesdropping. More sophisticated attacks consisted of UAV recorded video attacks, man-in-the-middle attacks, and Traffic Alert and Collision Avoidance System (TCAS) attacks. Countermeasures focusing on the categories of prevention, detection, and mitigation were discussed. Researched methods included stricter access control to UAV systems, detection of anomalies in the UAV sensor readings, and neutralization of attacks by countering the attack (i.e. for jamming) or avoiding the attack (i.e. for spoofing) by leaving the targeted area. As more UAVs are launched into a common airspace, it becomes important to devise smart countermeasures to counteract the actions performed by bad actors. Although the countermeasures listed in this paper are mainly applicable for communication and transponder-based attacks, the idea remains for the entire system. With newer technologies, these methods incorporate advancements such as the use of efficient machine learning algorithms and secure cryptography to create more robust UAV communication networks. Overall, this paper provides a comprehensive summary of the historical context of UAV-related cyberattacks, countermeasures, and the evolving trends and future advancements in these respective areas.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no conflict of interest. They also declare that they have no competing interest.

DATA AVAILABILITY

No data was used for the research in the article.

REFERENCES

- [1] H. A. Hashim, "Exponentially stable observer-based controller for VTOL-UAVs without velocity measurements," *Int. J. Control*, vol. 96, no. 8, pp. 1946–1960, Aug. 2023.
- [2] S. Bang, H. Kim, and H. Kim, "UAV-based automatic generation of high-resolution panorama at a construction site with a focus on preprocessing for image stitching," *Autom. Construct.*, vol. 84, pp. 70–80, Dec. 2017.
- [3] A. Ramachandran and A. K. Sangaiah, "A review on object detection in unmanned aerial vehicle surveillance," *Int. J. Cognit. Comput. Eng.*, vol. 2, pp. 215–228, Jun. 2021.
- [4] H. A. Hashim, A. E. E. Eltoukhy, and K. G. Vamvoudakis, "UWB ranging and IMU data fusion: Overview and nonlinear stochastic filter for inertial navigation," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 359–369, Jan. 2024.
- [5] J. Gu, T. Su, Q. Wang, X. Du, and M. Guizani, "Multiple moving targets surveillance based on a cooperative network for multi-UAV," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 82–89, Apr. 2018.
- [6] Y. Lim, A. Gardi, R. Sabatini, S. Ramasamy, T. Kistan, N. Ezer, J. Vince, and R. Bolia, "Avionics human-machine interfaces and interactions for manned and unmanned aircraft," *Prog. Aerosp. Sci.*, vol. 102, pp. 1–46, Oct. 2018.
- [7] D. Wanner, H. A. Hashim, S. Srivastava, and A. Steinhauer, "UAV avionics safety, certification, accidents, redundancy, integrity, and reliability: A comprehensive review and future trends," *Drone Syst. Appl.*, vol. 12, pp. 1–23, Jan. 2024.
- [8] T. Khamvilai, J. Dunham, E. Feron, and E. N. Johnson, "Avionics of aerial robots," *Current Robot. Rep.*, vol. 2, no. 2, pp. 113–124, Jun. 2021.
- [9] K. Hartmann and C. Steup, "The vulnerability of UAVs to cyber attacks—An approach to the risk assessment," in *Proc. 5th Int. Conf. Cyber Conflict (CYCON)*, Jun. 2013, pp. 1–23.
- [10] R. Mitchell and I.-R. Chen, "Adaptive intrusion detection of malicious unmanned air vehicles using behavior rule specifications," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 44, no. 5, pp. 593–604, May 2014.
- [11] D. T. Kuehl, *The Radar Eye Blinded: The USAF and Electronic Warfare 1945–1955*. Durham, NC, USA: Duke Univ., 1992.
- [12] V. P. Riabukha, "Radar surveillance of unmanned aerial vehicles," *Radioelectronics Commun. Syst.*, vol. 63, no. 11, pp. 561–573, 2020.
- [13] C. Rani, H. Modares, R. Sriram, D. Mikulski, and F. L. Lewis, "Security of unmanned aerial vehicle systems against cyber-physical attacks," *J. Defense Model. Simul., Appl., Methodol., Technol.*, vol. 13, no. 3, pp. 331–342, Jul. 2016.
- [14] Z. Lv, Y. Li, J. Wu, and H. Lv, "Securing the Internet of Drones against cyber-physical attacks," *IEEE Internet Things Mag.*, vol. 4, no. 4, pp. 74–78, Dec. 2021.
- [15] M. Leonardi, M. Strohmeier, and V. Lenders, "On jamming attacks in crowdsourced air traffic surveillance," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 36, no. 6, pp. 44–54, Jun. 2021.
- [16] A. Rugo, C. A. Ardagna, and N. E. Ioini, "A security review in the UAVNet era: Threats, countermeasures, and gap analysis," *ACM Comput. Surv.*, vol. 55, no. 1, pp. 1–35, Jan. 2023.
- [17] Q. Xiao, Y. Li, F. Luo, and H. Liu, "Analysis and assessment of risks to public safety from unmanned aerial vehicles using fault tree analysis and Bayesian network," *Technol. Soc.*, vol. 73, May 2023, Art. no. 102229.
- [18] T. E. Humphreys, "Detection strategy for cryptographic GNSS anti-spoofing," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 49, no. 2, pp. 1073–1090, Apr. 2013.
- [19] A. Jafarnia-Jahromi, A. Broumandan, J. Nielsen, and G. Lachapelle, "GPS vulnerability to spoofing threats and a review of anti-spoofing techniques," *Int. J. Navigat. Observ.*, vol. 2012, pp. 1–16, Jul. 2012.
- [20] H. J. Jo and W. Choi, "A survey of attacks on controller area networks and corresponding countermeasures," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6123–6141, Jul. 2022.
- [21] T. Li, J. Ye, J. Dai, H. Lei, W. Yang, G. Pan, and Y. Chen, "Secure UAV-to-vehicle communications," *IEEE Trans. Commun.*, vol. 69, no. 8, pp. 5381–5393, Aug. 2021.
- [22] W. BinSaeedan, A. Aldawsari, L. Alhussain, L. Alrushud, and L. Alfawzan, "Security challenges for UAV systems communications: Potential attacks and countermeasures," in *Unmanned Aerial Vehicles Applications: Challenges and Trends*. Cham, Switzerland: Springer, 2023, pp. 269–288.
- [23] B. Zohuri and B. Zohuri, "Electronic countermeasure and electronic counter-countermeasure," in *Radar Energy Warfare Challenges Stealth Technology*. Berlin, Germany: Springer, 2020, pp. 111–145.

- [24] H. R. Sonawane and S. P. Mahulikar, "Tactical air warfare: Generic model for aircraft susceptibility to infrared guided missiles," *Aerosp. Sci. Technol.*, vol. 15, no. 4, pp. 249–260, Jun. 2011.
- [25] F. Neri, *Introduction to Electronic Defense Systems*. Rijeka, Croatia: SciTech, 2006.
- [26] L. Wang, Y. Chen, P. Wang, and Z. Yan, "Security threats and countermeasures of unmanned aerial vehicle communications," *IEEE Commun. Standards Mag.*, vol. 5, no. 4, pp. 41–47, Dec. 2021.
- [27] H. Sedjelmaci, S. M. Senouci, and M.-A. Messous, "How to detect cyber-attacks in unmanned aerial vehicles network?" in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2016, pp. 1–6.
- [28] R. Shrestha, A. Omidkar, S. A. Roudi, R. Abbas, and S. Kim, "Machine-learning-enabled intrusion detection system for cellular connected UAV networks," *Electronics*, vol. 10, no. 13, p. 1549, Jun. 2021.
- [29] L. Wu and X. Cao, "Geo-location estimation from two shadow trajectories," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 585–590.
- [30] X. Li, W. Huangfu, X. Xu, J. Huo, and K. Long, "Secure offloading with adversarial multi-agent reinforcement learning against intelligent eavesdroppers in UAV-enabled mobile edge computing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 13914–13928, Dec. 2024.
- [31] H. Li, X. Li, F. Dunkin, Z. Zhang, and X. Lu, "Adaptive multi-granularity trust management scheme for UAV visual sensor security under adversarial attacks," *Comput. Secur.*, vol. 148, Jan. 2025, Art. no. 104108.
- [32] H. A. Hashim, "Advances in UAV avionics systems architecture, classification and integration: A comprehensive review and future perspectives," *Results Eng.*, vol. 25, Mar. 2025, Art. no. 103786.
- [33] H. A. Hashim, A. E. E. Eltoukhy, and A. Odry, "Observer-based controller for VTOL-UAVs tracking using direct vision-aided inertial navigation measurements," *ISA Trans.*, vol. 137, pp. 133–143, Jun. 2023.
- [34] A. Mpitzopoulos, D. Gavalas, C. Konstantopoulos, and G. Pantziou, "A survey on jamming attacks and countermeasures in WSNs," *IEEE Commun. Surveys Tuts.*, vol. 11, no. 4, pp. 42–56, 4th Quart., 2009.
- [35] H. Pirayesh and H. Zeng, "Jamming attacks and anti-jamming strategies in wireless networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 767–809, 2nd Quart., 2022.
- [36] K. Belwafi, R. Alkadi, S. A. Alameri, H. A. Hamadi, and A. Shoufan, "Unmanned aerial vehicles' remote identification: A tutorial and survey," *IEEE Access*, vol. 10, pp. 87577–87601, 2022.
- [37] Y. Zhi, Z. Fu, X. Sun, and J. Yu, "Security and privacy issues of UAV: A survey," *Mobile Netw. Appl.*, vol. 25, no. 1, pp. 95–101, Feb. 2020.
- [38] Z. Wu, T. Shang, and A. Guo, "Security issues in automatic dependent surveillance-broadcast (ADS-B): A survey," *IEEE Access*, vol. 8, pp. 122147–122167, 2020.
- [39] V. Hassija, V. Chamola, A. Agrawal, A. Goyal, N. C. Luong, D. Niyato, F. R. Yu, and M. Guizani, "Fast, reliable, and secure drone communication: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 4, pp. 2802–2832, 4th Quart., 2021.
- [40] Z. Wang, Y. Li, S. Wu, Y. Zhou, L. Yang, Y. Xu, T. Zhang, and Q. Pan, "A survey on cybersecurity attacks and defenses for unmanned aerial systems," *J. Syst. Archit.*, vol. 138, May 2023, Art. no. 102870.
- [41] A. E. Omolara, M. Alawida, and O. I. Abiodun, "Drone cybersecurity issues, solutions, trend insights and future perspectives: A survey," *Neural Comput. Appl.*, vol. 35, no. 31, pp. 23063–23101, Nov. 2023.
- [42] R. Aissaoui, J.-C. Deneuville, C. Guerber, and A. Pirovano, "A survey on cryptographic methods to secure communications for UAV traffic management," *Veh. Commun.*, vol. 44, Dec. 2023, Art. no. 100661.
- [43] A. Shafique, A. Mehmood, and M. Elhadeif, "Survey of security protocols and vulnerabilities in unmanned aerial vehicles," *IEEE Access*, vol. 9, pp. 46927–46948, 2021.
- [44] I. Svyd, I. Obod, and O. Maltsev, "Interference immunity assessment identification friend or foe systems," in *Data-Centric Business and Applications: ICT Systems—Theory, Radio-Electronics, Information Technologies and Cybersecurity*. Cham, Switzerland: Springer, 2021, pp. 287–306.
- [45] W. H. Harman, "TCAS—A system for preventing midair collisions," *Lincoln Lab. J.*, vol. 2, no. 3, pp. 437–457, 1989.
- [46] *Introduction to TCAS II*, U.S. Dept. Transp., Federal Aviation Admin., Washington, DC, USA, 2011, pp. 1–40, vol. 1, no. 1.
- [47] J. Sun, *The 1090 Megahertz Riddle: A Guide to Decoding Mode S and ADS-B Signals*, vol. 2. Delft The Netherlands: TU Delft OPEN Publishing, 2021, pp. 1–20.
- [48] S. J. Undertaking, "European drones outlook study: Unlocking the value for Europe," Tech. Rep., 2017.
- [49] A. Schelle, F. Völkl, R. T. Schwarz, A. Knopp, and P. Stütz, "Evaluation of a multi-mode-transceiver for enhanced UAV visibility and connectivity in mixed ATM/UTM contexts," *Drones*, vol. 6, no. 4, p. 80, Mar. 2022.
- [50] T. Karachalios, C. Xouris, and T. Orphanoudakis, "Optimizing UAV location awareness telemetry data for low power wide area network," in *Proc. 25th Euromicro Conf. Digit. Syst. Design (DSD)*, Aug. 2022, pp. 885–888.
- [51] P. Tedeschi, S. Sciancalepore, and R. Di Pietro, "ARID: Anonymous remote Identification of unmanned aerial vehicles," in *Proc. Annu. Comput. Secur. Appl. Conf.*, Dec. 2021, pp. 207–218.
- [52] B. Ly and R. Ly, "Cybersecurity in unmanned aerial vehicles (UAVs)," *J. Cyber Secur. Technol.*, vol. 5, no. 2, pp. 120–137, Apr. 2021.
- [53] I. A. Shah, "Privacy and security challenges in unmanned aerial vehicles (UAVs)," in *Cybersecurity in the Transportation Industry (UAVs)*. Berlin, Germany: Springer, 2024, pp. 93–115.
- [54] W. I. Alluhybi and O. H. Alhazmi, "Towards a threat model for unmanned aerial vehicles," in *Proc. ICTIDS*. Cham, Switzerland: Springer, Jan. 2021, pp. 319–328.
- [55] S. Rudys, J. Aleksandravicius, R. Aleksiejunas, A. Konovaltsev, C. Zhu, and L. Greda, "Physical layer protection for ADS-B against spoofing and jamming," *Int. J. Crit. Infrastruct. Protection*, vol. 38, Sep. 2022, Art. no. 100555.
- [56] A. C. Pogaku, D.-T. Do, B. M. Lee, and N. D. Nguyen, "UAV-assisted RIS for future wireless communications: A survey on optimization and performance analysis," *IEEE Access*, vol. 10, pp. 16320–16336, 2022.
- [57] T. Li, B. Wang, F. Shang, J. Tian, and K. Cao, "Threat model and construction strategy on ADS-B attack data," *IET Inf. Secur.*, vol. 14, no. 5, pp. 542–552, Sep. 2020.
- [58] B. S. Ali, W. Y. Ochieng, and R. Zainudin, "An analysis and model for automatic dependent surveillance broadcast (ADS-B) continuity," *GPS Solutions*, vol. 21, no. 4, pp. 1841–1854, Oct. 2017.
- [59] X. Prats, L. Delgado, J. Ramírez, P. Royo, and E. Pastor, "Requirements, issues, and challenges for sense and avoid in unmanned aircraft systems," *J. Aircr.*, vol. 49, no. 3, pp. 677–687, May 2012.
- [60] C. E. Lin and Y.-H. Lai, "Quasi-ADS-B based UAV conflict detection and resolution to manned aircraft," *J. Electr. Comput. Eng.*, vol. 2015, no. 1, pp. 1–12, 2015.
- [61] O. Fonseca, I. Cunha, E. Fazzion, W. Meira, B. A. d. Silva, R. A. Ferreira, and E. Katz-Bassett, "Identifying networks vulnerable to IP spoofing," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 3, pp. 3170–3183, Sep. 2021.
- [62] P.-Y. Kong, "UAV-assisted quantum key distribution for secure communications with resource limited devices," *IEEE Trans. Veh. Technol.*, vol. 73, no. 8, pp. 11923–11933, Aug. 2024.
- [63] H. Lee, S. Eom, J. Park, and I. Lee, "UAV-aided secure communications with cooperative jamming," *IEEE Trans. Veh. Technol.*, vol. 67, no. 10, pp. 9385–9392, Oct. 2018.
- [64] V. Chamola, P. Kotesch, A. Agarwal, N. Gupta, and M. Guizani, "A comprehensive review of unmanned aerial vehicle attacks and neutralization techniques," *Ad Hoc Netw.*, vol. 111, Feb. 2021, Art. no. 102324.
- [65] O. Westerlund and R. Asif, "Drone hacking with raspberry-pi 3 and WiFi pineapple: Security and privacy threats for the Internet-of-Things," in *Proc. 1st Int. Conf. Unmanned Vehicle Syst.-Oman (UVS)*, Feb. 2019, pp. 1–10.
- [66] G. de Carvalho Bertoli, L. A. Pereira, and O. Saotome, "Classification of denial of service attacks on Wi-Fi-based unmanned aerial vehicle," in *Proc. 10th Latin-American Symp. Dependable Comput. (LADC)*, Nov. 2021, pp. 1–6.
- [67] H. Sedjelmaci, S. M. Senouci, and N. Ansari, "A hierarchical detection and response system to enhance security against lethal cyber-attacks in UAV networks," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 48, no. 9, pp. 1594–1606, Sep. 2018.
- [68] D. Hawashin, M. Nemer, S. A. Gebreab, K. Salah, R. Jayaraman, M. K. Khan, and E. Damiani, "Blockchain applications in UAV industry: Review, opportunities, and challenges," *J. Netw. Comput. Appl.*, vol. 230, Oct. 2024, Art. no. 103932.
- [69] M. Mynuddin, S. U. Khan, R. Ahmari, L. Landivar, M. N. Mahmoud, and A. Homaifar, "Trojan attack and defense for deep learning-based navigation systems of unmanned aerial vehicles," *IEEE Access*, vol. 12, pp. 89887–89907, 2024.

- [70] J. Gaspar, R. Ferreira, P. Sebastião, and N. Souto, "Capture of UAVs through GPS spoofing using low-cost SDR platforms," *Wireless Pers. Commun.*, vol. 115, no. 4, pp. 2729–2754, Dec. 2020.
- [71] S. A. H. Mohsan, N. Q. H. Othman, Y. Li, M. H. Alsharif, and M. A. Khan, "Unmanned aerial vehicles (UAVs): Practical aspects, applications, open challenges, security issues, and future trends," *Intell. Service Robot.*, vol. 16, no. 1, pp. 109–137, Mar. 2023.
- [72] F. Tlili, L. C. Fourati, S. Ayed, and B. Ouni, "Investigation on vulnerabilities, threats and attacks prohibiting UAVs charging and depleting UAVs batteries: Assessments & countermeasures," *Ad Hoc Netw.*, vol. 129, Apr. 2022, Art. no. 102805.
- [73] M. Robinson, K. Jones, and H. Janicke, "Cyber warfare: Issues and challenges," *Comput. Secur.*, vol. 49, pp. 70–94, Mar. 2015.
- [74] D. P. Srirangam, K. Hemalatha, A. Vajravelu, and N. A. Kumar, "Safety and security issues in employing drones," in *Wireless Networks: Cyber Security Threats and Countermeasures*. Cham, Switzerland: Springer, 2023, pp. 103–131.
- [75] N. Vanitha and S. K. R. Selva, "Cyber security investigation of GPS-spoofing attack in military UAV networks," in *Proc. Int. Conf. Internet Everything Quantum Inf. Process*. Cham, Switzerland: Springer, Jan. 2024, pp. 116–124.
- [76] S. Malaviya, "Digitising the virtual: Movement and relations in drone warfare," *Millennium: J. Int. Stud.*, vol. 49, no. 1, pp. 80–104, Sep. 2020.
- [77] N. Taneski, B. Caminski, and A. Petrovski, "Use of weaponized unmanned aerial vehicles (UAVs) supported by gis as a growing terrorist threat," in *Science and Society Contribution of Humanities and Social Sciences*, 2020, pp. 553–567.
- [78] J. Zhao, X. Chao, Z. Chang, M. Cheriet, M. Kadoch, R. Liu, J. Lu, Y. Liu, and T. Hong, "Design of environmental backscatter tag antenna for 5G Internet of Things," in *Proc. IEEE Int. Symp. Broadband Multimedia Syst. Broadcast. (BMSB)*, Oct. 2020, pp. 1–4.
- [79] M. A. Alahe, L. Wei, Y. Chang, S. R. Gummi, J. Kemesi, X. Yang, K. Won, and M. Sher, "Cyber security in smart agriculture: Threat types, current status, and future trends," *Comput. Electron. Agricult.*, vol. 226, Nov. 2024, Art. no. 109401.
- [80] Y. Guo, M. Wu, K. Tang, J. Tie, and X. Li, "Covert spoofing algorithm of UAV based on GPS/INS-integrated navigation," *IEEE Trans. Veh. Technol.*, vol. 68, no. 7, pp. 6557–6564, Jul. 2019.
- [81] C. A. Roma, C. A. Tai, and M. A. Hasan, "Energy efficiency analysis of post-quantum cryptographic algorithms," *IEEE Access*, vol. 9, pp. 71295–71317, 2021.
- [82] D. He, S. Chan, and M. Guizani, "Communication security of unmanned aerial vehicles," *IEEE Wireless Commun.*, vol. 24, no. 4, pp. 134–139, Aug. 2017.
- [83] C. Popper, M. Strasser, and S. Capkun, "Anti-jamming broadcast communication using uncoordinated spread spectrum techniques," *IEEE J. Sel. Areas Commun.*, vol. 28, no. 5, pp. 703–715, Jun. 2010.
- [84] X. Zhang, H. Luo, K. Li, and O. Kaynak, "Time-domain frequency estimation with application to fault diagnosis of the unmanned aerial vehicles' blade damage," *IEEE Trans. Ind. Electron.*, vol. 69, no. 5, pp. 5257–5266, May 2022.
- [85] L. Meng, L. Yang, S. Ren, G. Tang, L. Zhang, F. Yang, and W. Yang, "An approach of linear regression-based UAV GPS spoofing detection," *Wireless Commun. Mobile Comput.*, vol. 2021, no. 1, Jan. 2021, Art. no. 5517500.
- [86] C. Ma, J. Yang, J. Chen, and C. Zhou, "Path following identification of unmanned aerial vehicles for navigation spoofing and its application," *ISA Trans.*, vol. 108, pp. 393–405, Feb. 2021.
- [87] Y. Wang, Z. Su, A. Benslimane, Q. Xu, M. Dai, and R. Li, "Collaborative honeypot defense in UAV networks: A learning-based game approach," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1963–1978, 2024.
- [88] A. S. Abdalla, K. Powell, V. Marojevic, and G. Geraci, "UAV-assisted attack prevention, detection, and recovery of 5G networks," *IEEE Wireless Commun.*, vol. 27, no. 4, pp. 40–47, Aug. 2020.
- [89] K. H. Terkildsen, U. P. Schultz, and K. Jensen, "Safely flying BVLOS in the EU with an unreliable UAS," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2021, pp. 591–601.
- [90] N. d'Ambrosio, G. Perrone, S. P. Romano, and A. Urraro, "A cyber-resilient open architecture for drone control," *Comput. Secur.*, vol. 150, Mar. 2025, Art. no. 104205.
- [91] R. C. J. Alexandre, L. E. G. Martins, and T. Gorschek, "Cybersecurity risk assessment for medium-risk drones: A systematic literature review," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 38, no. 6, pp. 28–43, Mar. 2023.
- [92] B. Branco, J. S. S. Silva, and M. Correia, "Cyber attacks on commercial drones: A review," *IEEE Access*, vol. 13, pp. 9566–9577, 2025.
- [93] H. A. Hashim, M. Abouheaf, and M. A. Abido, "Geometric stochastic filter with guaranteed performance for autonomous navigation based on IMU and feature sensor fusion," *Control Eng. Pract.*, vol. 116, Nov. 2021, Art. no. 104926.
- [94] N. El-Sheimy and A. Youssef, "Inertial sensors technologies for navigation applications: State of the art and future trends," *Satell. Navigat.*, vol. 1, no. 1, p. 2, Dec. 2020.
- [95] H. A. Hashim, "GPS-denied navigation: Attitude, position, linear velocity, and gravity estimation with nonlinear stochastic observer," in *Proc. Amer. Control Conf. (ACC)*, May 2021, pp. 1149–1154.
- [96] H. A. Hashim and A. E. E. Eltoukhy, "Landmark and IMU data fusion: Systematic convergence geometric nonlinear observer for SLAM and velocity bias," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3292–3301, Apr. 2022.
- [97] V. Gírbés-Juan, L. Armesto, D. Hernández-Ferrándiz, J. F. Dols, and A. Sala, "Asynchronous sensor fusion of GPS, IMU and CAN-based odometry for heavy-duty vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8617–8626, Sep. 2021.
- [98] Y. Zhang, "LILO: A novel LiDAR-IMU SLAM system with loop optimization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, pp. 2649–2659, Aug. 2022.
- [99] F. Betti Sorbelli, "UAV-based delivery systems: A systematic review, current trends, and research challenges," *ACM J. Auto. Transp. Syst.*, vol. 1, no. 3, pp. 1–40, Sep. 2024.
- [100] Y.-J. Tu and S. Piramuthu, "Security and privacy risks in drone-based last mile delivery," *Eur. J. Inf. Syst.*, vol. 33, no. 5, pp. 617–630, Sep. 2024.
- [101] R. L. Kumar, Q.-V. Pham, F. Khan, M. J. Piran, and K. Dev, "Blockchain for securing aerial communications: Potentials, solutions, and research directions," *Phys. Commun.*, vol. 47, Aug. 2021, Art. no. 101390.
- [102] K. Kim and Y. Kang, "Drone security module for UAV data encryption," in *Proc. Int. Conf. Inf. Commun. Technol. Conver. (ICTC)*, Oct. 2020, pp. 1672–1674.
- [103] D. He, S. Chan, and M. Guizani, "Drone-assisted public safety networks: The security aspect," *IEEE Commun. Mag.*, vol. 55, no. 8, pp. 218–223, Aug. 2017.
- [104] B. Semal, K. Markantonakis, and R. N. Akram, "A certificateless group authenticated key agreement protocol for secure communication in untrusted UAV networks," in *Proc. IEEE/AIAA 37th Digit. Avionics Syst. Conf. (DASC)*, Sep. 2018, pp. 1–8.
- [105] A. Y. Javaid, W. Sun, V. K. Devabhaktuni, and M. Alam, "Cyber security threat analysis and modeling of an unmanned aerial vehicle system," in *Proc. IEEE Conf. Technol. Homeland Secur. (HST)*, Nov. 2012, pp. 585–590.
- [106] A. Chauhan and M. R. Singla, "A detail review on unmanned aeronautical ad-hoc networks," *Int. J. Sci. Eng. Technol. Res.*, vol. 5, no. 5, pp. 1351–1360, 2016.
- [107] A. Ossamah, "Blockchain as a solution to drone cybersecurity," in *Proc. IEEE 6th World Forum Internet Things (WF-IoT)*, Jun. 2020, pp. 1–9.
- [108] V. Sokolov, P. Skladannyi, and N. Korshun, "ZigBee network resistance to jamming attacks," in *Proc. IEEE Int. Conf. Inf. Telecommun. Technol. Radio Electron. (UkrMiCo)*, Nov. 2023, pp. 161–165.
- [109] H. A. Hashim and K. G. Vamvoudakis, "Adaptive neural network stochastic-filter-based controller for attitude tracking with disturbance rejection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1217–1227, Jun. 2022.
- [110] H. J. Hadi, Y. Cao, S. Li, Y. Hu, J. Wang, and S. Wang, "Real-time collaborative intrusion detection system in UAV networks using deep learning," *IEEE Internet Things J.*, vol. 11, no. 20, pp. 33371–33391, Oct. 2024.
- [111] A. V. Jonnalagadda and H. A. Hashim, "SegNet: A segmented deep learning based convolutional neural network approach for drones wildfire detection," *Remote Sens. Appl., Soc. Environ.*, vol. 34, Apr. 2024, Art. no. 101181.



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